
Credit Risk

Fundamentals

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Chapter 1

Introduction

1.1 Objective

Credit risk represents one of the main exposures for financial intermediaries, defined as the possibility that a counterparty does not fulfill contractual obligations, with potential destabilizing effects both for the individual institution and for the financial system as a whole.

In the European context, increasing regulatory complexity and the evolution of operational practices require a structured approach that coherently integrates regulatory constraints and modeling methodologies. The availability of a harmonized framework helps reduce misalignment risks, fostering transparency and traceability in processes.

The overarching objective of the document is to provide a systematic introduction to credit risk management, bridging banking regulation and internal modeling within a coherent and up-to-date framework, in support of operational activities and control functions.

1.2 Scope of application

The scope of the analysis encompasses models and processes for measuring credit risk, European regulatory references and governance frameworks and controls, through a formal, shared language that ensures interpretive and operational consistency. Definitions, modeling assumptions and fundamental metrics are centralized.

In this context, two main approaches to measuring credit risk coexist, distinguished by purpose and scope of application:

- **A-IRB approach (Advanced Internal Ratings-Based):** provided for by Regulation (EU) No. 575/2013 (CRR) and the Basel standards, it is aimed at determining the minimum capital requirements to cover credit risk. It adopts a prudential approach, based on regulatory parameters such as probability of default (PD), loss given default (LGD), exposure at default (EAD) and maturity (M).
- **IFRS 9 approach:** introduced by international accounting standards, it is oriented toward the accounting representation of expected losses on financial tools (*Expected Credit Losses*). It is based on an economic-valuation approach, consistent with current market conditions and on Point-in-Time estimates, applied over a 12-month or lifetime horizon depending on the stage, useful for determining the carrying amount of credit assets.

The two approaches share the aim of measuring credit risk in a consistent and traceable way, but they address different objectives: A-IRB seeks to ensure capital soundness and regulatory

compliance, whereas IFRS 9 ensures the proper accounting representation of risk and expected losses in the group's financial statements.

1.3 A-IRB & IFRS 9 Scope

1.3.1 A-IRB Scope (CRR/Basel)

Under the European prudential framework (CRR), the *A-IRB* scope includes, for the purposes of the capital requirement for credit risk:

- **On-balance-sheet credit exposures:** they include all loans, mortgages and other credit assets that the bank has already granted and that are recorded on the balance sheet. They are divided into regulatory portfolios (for example corporates, households, public sector entities, banks) depending on the type of counterparty and the related risk profile.
- **Off-balance-sheet items:** they include commitments that the bank has undertaken but which have not yet resulted in an actual disbursement of funds, such as credit lines, guarantees, or commitments to finance a obligor in the future. To assess the risk of such items, the regulation uses conversion factors (*Credit Conversion Factors*) that estimate the portion of the amount potentially destined to turn into an actual exposure and classifies them into risk groups (*buckets*) based on the degree of enforceability of the commitment.
- **Derivatives and other financial transactions:** they include contracts such as swaps, options or futures, which generate counterparty credit risk (i.e. the possibility that the other party does not fulfill its obligations). These tools estimate the potential future exposure for counterparty credit risk (CCR), expressed as *Exposure at Default* (EAD) under the regulatory methodologies.

The *A-IRB* scope follows **prudential consolidation**, that is, it includes all entities belonging to the banking group for supervisory purposes (banks, financial intermediaries, securitization vehicles prudentially consolidated), even if not included in the statutory consolidated financial statements. Conversely, entities not material for regulatory purposes (i.e. real estate or industrial subsidiaries) are excluded.

1.3.2 IFRS 9 Scope (accounting)

For the purpose of recognizing *Expected Credit Losses* (ECL), the *IFRS 9* scope includes:

- **Financial assets:** they include loans and other financial tools held by the bank and measured according to two main criteria under IFRS 9:
 - at *amortized cost*, i.e. at the original value adjusted for repayments, accrued interest and expected losses;
 - at *fair value through other comprehensive income (FVOCI)*, i.e. at the updated market value, with unrealized gains or losses not immediately recognized in profit or loss but in a specific equity reserve.

Both categories include loans, mortgages and other credit exposures in the financial statements.

- **Irrevocable commitments and financial guarantees:** they include credit lines and guarantee contracts for which the bank has a contractual obligation to disburse funds or to respond in the event of obligor default. They are included in the scope only if *irrevocable*, that is, if the bank cannot revoke or reduce them freely without notice.

- **Impaired exposures (Stage 3):** refer to loans for which a loss event has already occurred, such as insolvency or debtor default. For such positions, the expected loss is calculated over the entire remaining life of the loan (*lifetime Expected Credit Loss*).

The scope of application of *IFRS 9* follows the principle of **accounting consolidation** established by IAS/IFRS standards. This means that risk assessment and recognition of expected losses are carried out for all entities included in the group's consolidated financial statements, regardless of whether or not they are subject to prudential requirements for supervisory purposes.

1.3.3 Essential differences in scope

Although the *A-IRB* and *IFRS 9* models both refer to the bank's credit exposures, they serve different purposes and are based on different inclusion criteria. In practice, this means that some items may fall within the A-IRB scope but not within the IFRS 9 scope (for example, a supervised special purpose entity that is not consolidated in statutory financial statements, or FVTPL trading tools). To ensure consistency between the two representations, banks therefore perform **formal reconciliations** between accounting and prudential data, so as to ensure that scope differences are tracked, explained and consistent.

1.4 New Definition of Default (NDoD)

The **New Definition of Default (NDoD)**, introduced under the initiative of the **European Banking Authority (EBA)**, came into force on January 1, 2021 (materiality thresholds were implemented by December 31, 2020). Its primary objective is to ensure greater *consistency, comparability and transparency* in the classification of non-performing credit exposures. This definition replaces previously heterogeneous national approaches by establishing common criteria and uniform thresholds for identifying default events, in accordance with **Article 178 of Regulation (EU) No. 575/2013 (CRR)**. The materiality threshold is set by Commission Delegated Regulation (EU) 2018/171, while the detailed application is specified in EBA/GL/2016/07. The NDoD applies both to IRB portfolios and standardised approaches, having a significant impact on **credit risk management processes** and on **regulatory reporting**.

1.4.1 Default Conditions: Criteria and Thresholds

According to the new framework, an exposure is considered to be in **default** when at least one of the following conditions is met:

- **More than 90 consecutive days past due:** The obligor has a payment past due on a credit obligation for a period exceeding 90 consecutive days, with the relevant *materiality thresholds* defined as follows:
 - **Retail obligors (individuals and SMEs):**
 - Absolute threshold: at least 100 euros (specific threshold for Italy).
 - Relative threshold: greater than 1% of the total exposure to the obligor.
 - **Non-retail obligors (corporates):**
 - Absolute threshold: at least 500 euros (specific threshold for Italy).
 - Relative threshold: greater than 1%.

The 90-day count begins only when both the absolute and relative thresholds are exceeded.

- **Unlikely to Pay (UTP):** The institution considers that the obligor is unlikely to fully meet its credit obligations without the use of actions such as the realisation of collateral or other credit protection mechanisms (CRM).

1.4.2 Additional Relevant Provisions

In addition to the above criteria, the NDoD introduces several important provisions:

- **Contagion effect:** If a debtor is in default on a significant exposure, the entire position of that obligor may be classified as defaulted, regardless of the performance of other exposures (*obligatory aggregation at obligor level*).
- **Minimum time in default (*cure period*):** Even when the circumstances that triggered the default cease to exist, the obligor must remain in default status for a minimum period of 90 days, extended to 12 months in case of distressed restructuring, except in duly documented exceptional cases.
- **Group-level default:** The default classification may be extended to all exposures related to entities legally or financially connected to the main obligor (cross-guarantees, control or financial links). Formally, if $i \in \mathcal{G}(k)$ represents an exposure belonging to group k and any exposure $j \in \mathcal{G}(k)$ is in default, then:

$$\mathbb{1}_{\text{DoD}}^{(i)}(t) = \max \left\{ \mathbb{1}_{\text{DoD}}^{(j)}(t) \mid j \in \mathcal{G}(k) \right\}.$$

The institution must ensure consistency across all exposures to the same debtor; for groups of connected obligors, the assessment of any contagion effects is carried out on a case-by-case basis, based on evidence of unlikely payment.

- **Backdating of the default event:** The formal default date may be backdated to the earliest point at which the default conditions were met:

$$t_{\text{DoD}} = \min \left\{ t \leq t_{\text{assessment}} \mid \text{default conditions satisfied} \right\}.$$

This principle ensures temporal consistency in historical time series and enables accurate backtesting of internal models. It must be supported by a proper *time-stamping* system and an audit trail ensuring full traceability of the event's origin.

1.4.3 Interaction with NPE and Forbearance

The NDoD is aligned with the EBA guidelines on:

- **Forbearance:** The granting of contractual modifications in response to a debtor's financial difficulties. Such exposures are subject to a cure period and intensive monitoring.
- **Non-Performing Exposures (NPE):** A harmonized classification framework that incorporates default events, forbearance measures and other indicators of credit deterioration.

Forborne exposures may be classified as performing or non-performing depending on the credit quality and recovery expectations. However, the existence of a forbearance measure may act as a triggering factor in the assessment of *unlikely to pay*.

1.5 Technical glossary

1.5.1 Fundamental parameters

The core parameters for measuring credit risk constitute the methodological basis on which both regulatory and accounting metrics are founded. They are estimated through quantitative models and validated according to prudential requirements, in order to ensure consistency, robustness and comparability across institutions.

- **PD (Probability of Default)**: represents the probability that a counterparty defaults within a given time horizon, typically one year. PD is the parameter that summarizes the borrower's credit quality and it is the cornerstone of internal rating models. An accurate PD estimate allows portfolios to be differentiated by risk grade and capital absorption to be calibrated proportionally. Its determination is carried out through statistical analyses based on historical default series and forward-looking risk indicators.
- **LGD (Loss Given Default)**: measures the percentage of exposure that the intermediary is unable to recover in the event of the counterparty's default. It includes the effects of collateral and personal guarantees, recovery costs and time to recovery/realisation. LGD therefore reflects the severity of loss and depends both on contractual characteristics and on the macroeconomic context. A prudent estimate of LGD makes it possible to assess the soundness of credit risk coverage and to incorporate stress scenarios into internal models.
- **EAD (Exposure at Default)**: represents the amount of exposure expected at the time of default. For revolving or granted facilities (for example, credit lines or revolving facilities), EAD is calculated by considering both the portion already drawn and the undrawn part converted through a conversion factor (Credit Conversion Factor, CCF). EAD is therefore the parameter that quantifies the base to which probabilities and loss rates are applied and it is crucial for correctly estimating expected loss.

The combination of PD, LGD and EAD makes it possible to translate credit risk into a monetary measure of expected loss. They constitute input parameters both in regulatory models (for the determination of RWA and capital requirements) and in accounting models under IFRS 9 (for the estimation of expected credit losses). In this sense, the three parameters represent a common language between prudential supervision and accounting standards, ensuring methodological consistency and transparency in assessments.

1.5.2 Conversion Factor (CCF)

The *Credit Conversion Factor* (CCF) is the parameter used to estimate the unused portion of a credit line that, at the time of the counterparty's default, is expected to be actually disbursed and thus converted into exposure. It makes it possible to account for the risk of *drawdown*, that is, the possibility that the borrower will use the credit reserves still available, when close to insolvency.

For the calculation of the exposure at default (EAD), the unused portion of a facility is therefore weighted by the CCF and added to the amount already used:

$$\text{EAD} = \text{Drawn} + \text{CCF} \times (\text{Granted} - \text{Drawn}).$$

The CCF is generally expressed as a percentage and varies according to the type of product (i.e. revolving credit lines, revocable credit facilities, trade guarantees), the contractual characteristics and historical statistical evidence. Higher CCF values reflect a greater likelihood that the lines will be drawn close to default.

1.5.3 Fundamental metrics

The measurement of credit risk is structured around three main metrics, each of which plays a distinctive role in prudential and accounting management.

- **Expected loss (EL)**: represents the average loss that the intermediary expects to incur over a given time horizon. It is calculated as the product of the probability of default (PD), the loss given default (LGD) and the exposure at default (EAD):

$$\text{EL} = \text{PD} \times \text{LGD} \times \text{EAD}.$$

EL therefore reflects the statistically predictable amount of loss and is already taken into account in pricing processes, margin allocation and the determination of value adjustments on loans.

- **Unexpected loss (UL):** measures the variability of actual losses around expected loss. It represents the potential deviation from average losses, estimated using prudential confidence intervals. UL constitutes the unpredictable component of losses and justifies the need for risk capital to safeguard the stability of the intermediary, forming the basis for the determination of regulatory capital requirements.
- **Expected Credit Loss (ECL, IFRS 9):** is the accounting measure of expected losses introduced by the International Financial Reporting Standard 9. Unlike regulatory expected loss, ECL applies to the measurement of loans on the balance sheet, considering two possible horizons: expected losses over 12 months (12-month ECL) and those over the tool's entire remaining life (lifetime ECL). ECL directly affects the income statement and equity through loss allowances for credit risk.

1.5.4 Risk-Weighted Assets (RWA)

Risk-Weighted Assets (RWA) represent the reference measure for determining the minimum regulatory capital required of intermediaries. RWA convert credit exposures into “weighted” values based on the risk profile (credit, market, operational), so as to express, in homogeneous terms, the amount of assets to which the capital ratios are applied. In the case of credit risk, RWA are determined through a function that integrates the main risk parameters:

$$K = f(\text{PD}, \text{LGD}, \text{EAD}, M),$$

where:

- PD: the counterparty's probability of default;
- LGD: the percentage loss in the event of default;
- EAD: the exposure at the time of default;
- M : the maturity, which is the effective remaining duration of the transaction.

The coefficient K measures the capital required to cover the unexpected loss associated with the credit portfolio. By applying K to the exposure, one obtains the amount of the corresponding RWA, which constitutes the basis for calculating the capital requirements.

The minimum regulatory capital requirement, established by the Basel Accords and implemented in European regulation, is set at 8% of total RWA. This percentage represents the minimum threshold that banks must maintain as supervisory capital, in order to ensure capital soundness and resilience to scenarios of unexpected loss.

1.5.5 Regulatory approaches

The regulatory framework provides for three alternative approaches to determining capital requirements for credit risk. They differ in the degree of methodological complexity, in the use of internal estimates and in the level of discretion granted to intermediaries.

- **Standardised Approach (STD):** is based on the application of risk weights predetermined by prudential regulation, differentiated according to the external rating assigned to counterparties by recognized agencies. In the absence of a rating, standard weights are

applied according to the type of counterparty (for example, banks, corporates, sovereigns). This approach has the advantage of simplicity and comparability, but it may be less sensitive to the actual risk profile of portfolios. The *Standardised* approach constitutes the common baseline for all intermediaries, especially those that are smaller or less complex.

- **Foundation Internal Ratings-Based (F-IRB):** entails the use of banks' internal estimates for the probability of default (PD) parameter, while the loss given default (LGD) and exposure at default (EAD) values are prescribed by the supervisory authority. This approach introduces greater risk sensitivity than the Standardised Approach, while maintaining a certain degree of regulatory uniformity. The goal is to support a gradual transition toward the use of internal models, while safeguarding comparability across institutions.
- **Advanced Internal Ratings-Based (A-IRB):** allows banks, subject to approval by the supervisory authority, to internally estimate all relevant parameters: PD, LGD and EAD. The advanced approach enables modeling that better reflects portfolio characteristics and greater efficiency in capital allocation, but it entails stringent requirements in terms of governance, data quality and model validation. In this context, the supervisory authority plays a crucial oversight role to ensure the robustness and reliability of internal estimates. The *A-IRB* approach targets the most advanced institutions, capable of developing sophisticated internal rating systems and meeting advanced requirements for control, validation and audit.

The coexistence of the three approaches reflects the need to balance implementation simplicity, comparability across banks and risk sensitivity.

Chapter 2

Data architecture and models

2.1 Estimation of PD (Probability of Default)

2.1.1 Data and sources

The estimation of the probability of default (PD) requires a rigorous data design ensuring completeness, temporal consistency and representativeness of the sample. The information sources are organized as follows.

Internal sources

- **Master data and consolidation perimeter:** unique identifiers, group structure, ATECO, NACE sector, country of residence, counterparty class (retail, SME, corporate).
- **Economic–financial statement data:** revenues, EBITDA, financial leverage, interest coverage, working capital and liquidity/solvency indicators, harmonized according to accounting standards.
- **Credit and behavioral performance:** utilization/approved amount, overdrafts, days past due, unpaid items, restructurings/forborne exposures, internal rating history, early warning events.
- **Collateral and mitigants:** type (secured/personal), appraisal values, loan-to-value ratios, internal haircuts (prudential discounts for managing deterioration risk), status and last update date.
- **Default event history:** date $t_{default}$, cause, procedural status, consistent with the internal/regulatory definition of default.

External sources

- **Credit bureaus and credit registers:** positions with other intermediaries, external performance behavior, adverse events, unpaid exposures; used for enrichment and consistency checks.
- **External ratings and market information:** when available, used for plausibility checks and benchmarking.
- **Macro–sectoral variables for Point in Time (*PiT*) calibrations:** cyclical indicators and sectoral proxies adopted for the cyclical anchoring of estimates.

Temporal design and consistency

The temporal design for PD estimation requires a clear separation between the period in which covariates are observed and that in which the outcome (default/non-default) is measured, as well as consistency of *time-stamping* across all data sources. We denote by t_0 the reference date for measuring the counterparty's risk profile.

- **Observation window of covariates.** Let $L > 0$ be the width (in months) of the covariate information window:

$$\mathcal{W}_X(t_0; L) = [t_0 - L, t_0] \text{ in } X,$$

within which the explanatory variables X (financial and economic, behavioral, credit bureau) are extracted with time-stamping consistent at t_0 . For each continuous covariate $x(t)$, a summary rule $g(\cdot)$ is specified (i.e. last available value, mean, quantile):

$$\tilde{x}(t_0; L) = g(\{x(t) : t \in \mathcal{W}_X(t_0; L)\}),$$

thus obtaining the input vector $\tilde{x}(t_0; L)$ free of any information posterior to t_0 .

- **Performance window of the outcome.** For a one-year PD, the performance horizon is fixed at $H = 12$ months:

$$\mathcal{W}_Y(t_0; H) = (t_0, t_0 + H] \text{ in } Y,$$

The target is defined over the horizon after t_0 , while covariates are built *ex-ante* with respect to t_0 .

- **Consistency of time-stamping and prevention of *label leakage*.** All data sources (master, management, accounting, collateral, credit registers) are aligned at t_0 through unique keys and consistent timestamps:
 - each variable $\tilde{x}_j$ is computed using only data with *timestamp* $\leq t_0$;
 - the outcome $Y(t_0; H)$ derives exclusively from events with *timestamp* in $\mathcal{W}_Y(t_0; H)$;
 - potential informational drifts (i.e. late accounting entries referring to periods after t_0) are avoided through *cut-off* rules and inter-system reconciliations.

This prevents the so-called *label leakage*, i.e. the improper use, during development, of information that reflects the future outcome.

- **Unique definition of default.** The definition of default is harmonized across all systems and applied ex ante in the construction of the *target*. Denoting by t_{DoD} the date of the first occurrence of default according to the adopted definition, the *label* is obtained as:

$$Y(t_0; H) = \begin{cases} 1, & \text{if } t_{\text{DoD}} \in \mathcal{W}_Y(t_0; H), \\ 0, & \text{otherwise,} \end{cases}$$

thus ensuring that rules, thresholds and materiality are unique and traceable.

- **Development, validation and *out-of-time* samples.** To assess the robustness and generalization of the model, the sample is split into:

$$\mathcal{D}_{\text{dev}} \cup \mathcal{D}_{\text{val}} \cup \mathcal{D}_{\text{oout}} = \mathcal{D},$$

where $\mathcal{D}_{\text{dev}}$ is used for estimation and variable selection, $\mathcal{D}_{\text{val}}$ for internal *in-sample* and *out-of-sample* validation over the same time horizon, and $\mathcal{D}_{\text{oout}}$ for *out-of-time* verification,

i.e. over a subsequent and non-overlapping period, so as to test the temporal stability of the relationship $(\tilde{\mathbf{x}}, Y)$. Operationally:

$$\mathcal{D}_{\text{oout}} = \{ (\tilde{\mathbf{x}}(t_0; L), Y(t_0; H)) : t_0 \in \mathcal{T}_{\text{oout}} \subset \mathbb{T}, \mathcal{T}_{\text{oout}} \cap \mathcal{T}_{\text{dev}} = \emptyset \},$$

where $\mathbb{T}$ is the total available time horizon, while $\mathcal{T}_{\text{dev}}$ and $\mathcal{T}_{\text{oout}}$ are disjoint time intervals, used to measure the discriminatory power and calibration consistency on a calendar different from the development one.

Summary scheme.

For a one-year PD horizon, the information flow is summarized as:

$$\underbrace{[t_0 - L, t_0]}_{\text{Observation window}} \xrightarrow{\tilde{\mathbf{x}}(t_0; L)} \underbrace{(t_0, t_0 + 12]}_{\text{Performance window}} \xrightarrow{Y(t_0; 12)} \text{Validation and calibration,}$$

ensuring that measurements are traceable, temporally consistent and replicable across development, validation and *out-of-time* sets.

Data quality, representativeness and data treatment

Once the development, validation and *out-of-time* samples are defined, data-quality checks and issue management are performed as follows:

- **Data-quality checks:** controls and checks of correctness across sources and inter-system reconciliations, using unique keys and deduplication processes.
- **Treatment of missing values and outliers:** application and documentation of pre-processing rules (imputation, *winsorization* - replacement of outliers with values closer to the sample mean, median, capping or flooring), ensuring traceability of choices and model impacts.
- **Stationarity and representativeness:** *drift* and *Population Stability* analyses to monitor distributional shifts between development and application phases.

2.1.2 Statistical models

In the context of PD estimation, the methodological framework includes binary-response statistical models, particularly logistic regression (*logit*) and *scorecards* derived from *Weight of Evidence* (WOE) transformations. Categorical variables (and, more generally, covariates discretized into *bins*) are often transformed into WOE to enhance model stability and interpretability. The development process follows an ordered path including univariate analysis, multicollinearity controls and multivariate estimation.

Univariate analysis and binning

Before multivariate estimation, each covariate is analyzed univariately to verify its ability to separate defaults from non-defaults and to define an informative and, where necessary, monotonic *binning* with respect to PD. Binning reduces noise, handles outliers and enables consistent WOE transformation.

Reduction of multicollinearity

Since highly correlated covariates may destabilize estimates (*inflated variances*) and blur coefficient interpretation, variable-selection and reduction procedures are applied: exclusion of redundant predictors, preference for parsimonious specifications and cross-correlation checks prior to joint estimation.

Weight of Evidence (WOE) transformation

For each variable (continuous or categorical) divided into B mutually exclusive bins $b = 1, \dots, B$, let G_b and E_b denote the counts of *good* and *bad* observations in bin b and $G = \sum_b G_b$, $E = \sum_b E_b$. The WOE of bin b is:

$$\text{WOE}_b = \ln\left(\frac{G_b/G}{E_b/E}\right),$$

i.e. the logarithm of the ratio between the share of non-defaults and the share of defaults in the bin. Original covariates are replaced by their respective WOE_b , yielding (quasi)–linear relationships in log-odds and improved temporal stability. In the presence of zero counts, additive minima (*smoothing*) or special-case values are introduced to avoid undefined ratios.

Scorecard as linear model in WOE

A *scorecard* is an additive score representation based on WOE:

$$\eta(\mathbf{x}) = \beta_0 + \sum_{j=1}^p \sum_{b \in \mathcal{B}_j} \beta_{j,b} \mathbb{1}\{x_j \in b\} \text{WOE}_{j,b},$$

where $\mathcal{B}_j$ is the bin partition of the j -th variable and $\eta(\mathbf{x})$ is the *linear predictor*, whose parameters are estimated by maximum likelihood maximization.

The final specification of variables is obtained by jointly estimating coefficients on a pre-selected set of covariates, ensuring sign consistency with economic-credit rationale, parameter stability and adequate discriminatory power.

Maximum likelihood estimation

Given n independent cases $(\mathbf{x}_i, y_i)$, the likelihood function is:

$$L(\boldsymbol{\beta}) = \prod_{i=1}^n [\pi(\mathbf{x}_i)]^{y_i} [1 - \pi(\mathbf{x}_i)]^{1-y_i}, \quad \ell(\boldsymbol{\beta}) = \sum_{i=1}^n \left\{ y_i \log \pi(\mathbf{x}_i) + (1 - y_i) \log [1 - \pi(\mathbf{x}_i)] \right\},$$

where π is the probability/density function. Parameters $\boldsymbol{\beta}$ are estimated by maximizing $\ell(\boldsymbol{\beta})$, which is the logarithm of $L(\boldsymbol{\beta})$.

Logit model: structure and interpretation

Let $Y \in \{0, 1\}$ be the outcome (default/non-default) observed over the performance horizon and $\mathbf{x} = (x_1, \dots, x_p)$ the vector of covariates computed at t_0 . Logistic regression specifies the conditional PD as:

$$\Pr(Y = 1 \mid \mathbf{x}) = \pi(\mathbf{x}) = \frac{1}{1 + \exp(-\eta(\mathbf{x}))}.$$

In terms of odds and log-odds:

$$\text{odds}(\mathbf{x}) = \frac{\pi(\mathbf{x})}{1 - \pi(\mathbf{x})}, \quad \log(\text{odds}(\mathbf{x})) = \eta(\mathbf{x}),$$

so that each coefficient β_j of the linear predictor represents the additive change in log-odds associated with a one-unit increase in x_j , holding other covariates constant.

2.1.3 Calibration

By calibration we mean the set of transformations that, starting from the raw model output (i.e. score or log/odds), produce a probability of default (PD) estimate that is consistent with empirical evidence and with the intended use (managerial or regulatory). Let $s = \eta(\mathbf{x})$ denote the score obtained from the statistical model estimated at time t_0 and let H be the performance horizon (for one-year PD, $H = 12$ months).

Point in Time (PiT) calibration

PiT PD incorporates the current state of the business cycle and the conjunctural conditions at time t_0 . A useful specification is:

$$\text{PD}_{\text{PiT}}(t_0; H \mid \mathbf{x}, \mathbf{z}_{t_0}) = \frac{1}{1 + \exp(a + b s + \gamma^\top \mathbf{z}_{t_0})},$$

where $\mathbf{z}_{t_0}$ collects macro and sectoral factors observed at t_0 (cycle, interest rates, sector indicators). The pair (a, b) performs the *anchoring* of the score to the realized evidence in the calibration sample, while γ transfers the cyclical effect into the probability. In aggregated form for score classes/deciles $g = 1, \dots, G$, PiT anchoring imposes the coherence constraint on observed rates:

$$\sum_{g=1}^G \widehat{\text{PD}}_{\text{PiT},g} N_g \approx \sum_{g=1}^G D_g, \quad \widehat{\text{PD}}_{\text{PiT},g} = \frac{1}{1 + \exp(a + b \bar{s}_g)},$$

where N_g is the number of exposures in group g , $\bar{s}_g$ the group's average score and D_g the number of defaults observed over the horizon H in group g . The system (a, b) is chosen to satisfy (on average) the empirical rate while preserving *monotonicity* ($\bar{s}_{g_1} < \bar{s}_{g_2} \Rightarrow \widehat{\text{PD}}_{\text{PiT},g_1} \leq \widehat{\text{PD}}_{\text{PiT},g_2}$).

Through the Cycle (TtC) calibration

TtC PD neutralizes the cyclical component as far as possible, yielding values *stabilized* with respect to the conjuncture. Conceptually:

$$\text{PD}_{\text{TtC}}(\mathbf{x}) = \mathbb{E}_{\mathbf{z}}[\text{PD}_{\text{PiT}}(t_0; H \mid \mathbf{x}, \mathbf{z})],$$

i.e. a (long-run) average of the *PiT* PD across business-cycle states $\mathbf{z}$. Operationally, TtC calibration is implemented by removing the effect of cyclical drivers (setting $\gamma = \mathbf{0}$ or fixing them at long-run average values) and recalibrating (a, b) to *medium/long-run* historical frequencies for homogeneous classes $(\tilde{a}, \tilde{b})$:

$$\sum_{g=1}^G \widehat{\text{PD}}_{\text{TtC},g} N_g \approx \sum_{g=1}^G \bar{D}_g, \quad \widehat{\text{PD}}_{\text{TtC},g} = \frac{1}{1 + \exp(\tilde{a} + \tilde{b} \bar{s}_g)},$$

with $\bar{D}_g$ the average annual number of defaults in group g over a multi-year window. The result is a PD surface that is *smoother* over time, suitable for uses requiring stability.

Long Run Average (LRA)

Long Run Average denotes the long-term historical average of default rates for homogeneous risk classes (ratings, score deciles, segments), estimated over a multi/year horizon spanning multiple cycle phases. For a group g :

$$\widehat{\text{PD}}_{\text{LRA},g} = \frac{\sum_{t=1}^T D_{g,t}}{\sum_{t=1}^T N_{g,t}} \quad (\text{frequency average}),$$

or

$$\widehat{\text{PD}}_{\text{LRA},g} = 1 - \prod_{t=1}^T (1 - d_{g,t})^{w_t} \quad (\text{geometric average}),$$

where $D_{g,t}$ and $N_{g,t}$ are, respectively, defaults and exposures of group g in year t , $d_{g,t} = D_{g,t}/N_{g,t}$ and w_t are weights (i.e. uniform). LRA provides a long-run *anchor* to which estimated PDs can be tied, ensuring consistency with historical targets.

Operational calibration tools

Regardless of the chosen perimeter (PiT, TtC, or LRA), the final mapping $\mathcal{C} : s \mapsto \widehat{\text{PD}}$ is determined by imposing coherence and monotonicity constraints. Commonly used tools include:

- *Logistic/Platt scaling*: $\widehat{\text{PD}}(s) = 1/(1+\exp(a+bs))$ with (a, b) selected to satisfy average-rate constraints and, where applicable, portfolio-level *targets*;
- *Isotonic regression*: a nonparametric, monotone estimator in s , useful when the score→PD relationship is not well approximated by a logistic function;
- *Binning* into risk classes g with *smoothing* (i.e. additive minima) to ensure robustness in rare classes.

In all cases, *mass balance* checks are adopted:

$$\sum_{i \in \mathcal{S}} \widehat{\text{PD}}_i \approx D_{\mathcal{S}}, \quad \forall \mathcal{S} \in \{\text{total portfolio, segments, classes}\},$$

i.e. the sum of calibrated PDs approximates the defaults observed over horizon H on the same perimeter.

Choice of approach and uses

The choice among *PiT*, *TtC* and *LRA* depends on the intended use:

- *PiT*: rapidly reflects current conditions; suitable for cycle-sensitive measures and short-term managerial applications;
- *TtC*: dampens cyclicalities, producing more stable profiles; useful when inter/temporal stability is a priority;
- *LRA*: anchors PDs to multi-year historical averages for homogeneous classes; indicated for long-term comparisons and stabilization objectives.

In practice, a *chain* of steps is often applied: model estimation → PiT calibration consistent with observed rates → transformation toward TtC/LRA for uses requiring lower cyclical sensitivity, while preserving monotonicity and level coherence.

2.1.4 Performance metrics

Evaluating the performance of a probability-of-default (PD) model requires measures of its *discriminatory power*, i.e. the ability to rank counterparties so that those that will default receive, on average, higher risk scores than sound positions. Reference indicators are the *Receiver Operating Characteristic* (ROC) with its *Area Under the Curve* (AUC), the *Cumulative Accuracy Profile* (CAP) and the *Accuracy Ratio* (AR).

Notation setup

Let $Y \in \{0, 1\}$ be the outcome at horizon H (default/non-default) and $S \in \mathbb{R}$ the model score (higher $S \Rightarrow$ higher risk). For a threshold τ , define:

$$\text{TPR}(\tau) = \Pr(S > \tau \mid Y = 1), \quad \text{FPR}(\tau) = \Pr(S > \tau \mid Y = 0),$$

where TPR is *sensitivity* (or *recall*), i.e. the percentage of predicted defaults relative to those observed, while FPR is the false-positive rate, i.e. the percentage of predicted defaults among counterparties that do not default.

ROC curve and AUC

The *ROC* is the locus of points $(\text{FPR}(\tau), \text{TPR}(\tau))$ as τ varies. A more discriminating model yields a ROC closer to the vertex $(0, 1)$. The *AUC* is the area underneath:

$$\text{AUC} = \int_0^1 \text{TPR}(\text{FPR}^{-1}(u)) \, du,$$

with $\text{AUC} \in [0, 1]$. A random model has $\text{AUC} \approx 0.5$; perfect ranking implies $\text{AUC} = 1$.

CAP curve

The *Cumulative Accuracy Profile* represents, for each cumulative share $q \in [0, 1]$ of the population ordered by decreasing score (from highest to lowest risk), the cumulative share of captured defaults:

$$\text{CAP}(q) = \Pr(S \text{ among the top } q \text{ quantiles} \mid Y = 1).$$

Three curves are relevant: (i) the *random* diagonal $\text{CAP}_{\text{rand}}(q) = q$ – the area under the diagonal, (ii) the model curve, (iii) the *perfect* curve, which captures all defaults in the first percentiles. The larger the distance between the model's CAP and the diagonal, the better the discrimination.

Accuracy Ratio (AR)

The *AR* normalizes the area between the model CAP and the random diagonal by the maximum attainable area (perfect curve). Let $\mathcal{A}(\cdot)$ denote the area under a curve:

$$\text{AR} = \frac{\mathcal{A}(\text{CAP}_{\text{mod}}) - \mathcal{A}(\text{CAP}_{\text{rand}})}{\mathcal{A}(\text{CAP}_{\text{perf}}) - \mathcal{A}(\text{CAP}_{\text{rand}})} \in [0, 1].$$

In practice, $\text{AR} = 0$ indicates no discriminatory power (random behavior), while $\text{AR} = 1$ corresponds to perfect ranking. In many implementations, with appropriate scaling conventions, AR is monotonically increasing with AUC.

Joint use of metrics

- *ROC* and *AUC* measure discrimination independently of default prevalence; they are robust to threshold choices.
- *CAP* and *AR* provide a cumulative view oriented to capturing defaults in the higher-risk tails, useful for cutoff policies and prioritization.
- Metrics should be computed on *validation* and *out-of-time* samples to assess robustness and temporal stability.

2.1.5 Stability and drift

The temporal stability of a probability of default (PD) model concerns the system's ability to maintain, *ceteris paribus*, a consistent ordering and risk levels over time. Two operational tools are used to detect distributional changes or model drift: the *Population Stability Index* (PSI) and *migration matrices* across risk classes/segments.

Population Stability Index (PSI)

PSI measures the deviation between the distribution of a variable (i.e. score or estimated PD) observed in a current period and the corresponding distribution in a reference period (development or validation). Consider a partition into B mutually exclusive intervals (or *bins*) $b = 1, \dots, B$. Let p_b be the relative frequency in the reference period and q_b the frequency observed in the current period, with $\sum_b p_b = \sum_b q_b = 1$. PSI is obtained by aggregating the contributions of each bin:

$$\text{PSI} = \sum_{b=1}^B (q_b - p_b) \ln\left(\frac{q_b}{p_b}\right),$$

with a standard *smoothing* convention to avoid undefined ratios in low-frequency bins. Values close to zero indicate stability; increasing values signal distributional *drift*. In practice, PSI is computed on:

- *score/PD*: to verify whether the current population exhibits a risk profile consistent with that of development;
- *key drivers*: to monitor the most relevant predictors and capture structural drifts in the data base.

Bin construction is consistent with the development *binning* (or stable percentile partitions) to ensure comparability and robustness. In the literature and banking practice, PSI is interpreted according to established empirical thresholds:

- $\text{PSI} < 0.10 \rightarrow$ **acceptable stability**: the current population is broadly consistent with the development or validation population.
- $0.10 \leq \text{PSI} < 0.25 \rightarrow$ **moderate change**: signals a drift to be analyzed, possibly due to shifts in obligor mix or macroeconomic shocks.
- $\text{PSI} \geq 0.25 \rightarrow$ **significant change**: indicates a possible loss of model validity and requires in-depth investigation or recalibration.

Migration matrices

Migration matrices describe transitions between risk classes/segments (i.e. *rating buckets* or score deciles) observed between two dates. Let $\mathcal{G} = \{1, \dots, G\}$ denote the set of classes ordered from lowest risk (1) to highest risk (G). The one-step transition matrix is

$$\mathbf{P} = [p_{ij}]_{i,j=1}^G, \quad p_{ij} = \Pr(\text{class}(t_1) = j \mid \text{class}(t_0) = i),$$

satisfying $p_{ij} \geq 0$ and $\sum_{j=1}^G p_{ij} = 1$ for every row i . Empirical observation of $\mathbf{P}$ over homogeneous horizons allows one to:

- measure *upgrades/downgrades* (migrations toward better/worse classes) and diagonal concentrations (p_{ii});
- highlight directional drifts (i.e. a systematic increase in movements toward high-risk tails);

- compare time cohorts ¹ to verify the stability of the assignment process and the ordering.

Over multiple horizons, the h -step transition is $\mathbf{P}^{(h)} = \mathbf{P}^h$ (Markov composition on homogeneous intervals), useful for time-consistency checks of the ordering induced by the model.

Joint interpretation and operational thresholds

PSI and migration matrices provide complementary readings:

- PSI quantifies the *global* deviation of distributions (score/PD or drivers);
- migrations show *where* the deviation is concentrated (which classes change and in which direction).

Diagnostics are completed by segment analyses (retail/SME/corporate portfolios), driver-level *drill-down* and *out-of-time* cohorts, maintaining consistency with observation/performance windows and with data *time-stamping*.

2.2 Estimation of LGD (Loss Given Default)

2.2.1 Data and definitions

Loss Given Default (LGD) is defined as one minus the *Recovery Rate* (RR):

$$\text{LGD} = 1 - \text{RR}.$$

This definition is the operational starting point for all subsequent measurement and modeling choices. RR is intended as the *share of exposure recovered net of costs*, properly discounted, relative to the exposure at default (EAD):

$$\text{RR} = \frac{\text{discounted recoveries} - \text{recovery costs}}{\text{EAD}}$$

so that the measure is consistent with the actual economic loss.

Information scope and granularity

Accurate estimation of **Loss Given Default (LGD)** requires a structured and comprehensive data set, built for each defaulted exposure and with time-stamping consistent with the default classification date. This data set must allow the detailed reconstruction of cash flows, timings and recovery conditions, thereby ensuring robustness, comparability and replicability of the measures.

Workout timelines

The time dimension is fundamental for estimating the actual value of recoveries:

- for *closed* positions, consider the **Time To Workout** (TTW), i.e. the interval between the default date and the final closure of the position;
- for *open* positions, use projections over a **Maximum Recovery Period** (MRP), defined in line with regulation and market practices.

The goal is to ensure that estimates reflect not only the amounts recovered but also the *timing* with which they materialize.

¹Cohorts are defined as groups of observations (e.g. customers or exposures) that share the same reference date t_0 and are tracked over a common time horizon, so that performance and default behaviour can be consistently compared across homogeneous time intervals.

Recovery cash flows (CF_t) and costs

Cash flows are the amounts collected after default. They include:

- voluntary repayments by the debtor;
- realisation of real collateral (*foreclosure sales*, judicial auctions, market sales);
- enforcement of personal guarantees (i.e. sureties, counter-guarantees).

Recovery costs must be explicitly subtracted from collected flows. They include:

- legal and court expenses;
- administrative and internal management costs;
- charges related to maintenance or management of collateral (i.e. real estate).

The loss rate in the event of default is therefore defined as:

$$LGD = 1 - RR = 1 - \frac{1}{EAD} \sum_{t=1}^T \left(\frac{CF_t}{(1+r)^t} - C_t \right)$$

where EAD is the exposure at default, r is the discount rate consistent with market curves and C_t are the recovery costs incurred at time t .

Homogeneity and data granularity

To ensure comparability, data must be organized into **homogeneous sets**, distinguishing:

- tool type and product (mortgages, leasing, revolving credit, unsecured loans);
- type of guarantee (real, personal, unsecured);
- geographical area and jurisdiction (differences in recovery times and rates);
- administrative and procedural status of the position (past due, UTP, or non-performing).

These elements are used to build homogeneous subsets, define LGD response functions and apply monotonicity or smoothing rules in calibration. Moreover, the observation horizon must be sufficiently long to capture the *cyclical*ity of recoveries and ensure statistical representativeness.

Collateral information

The assessment of guarantees is central to LGD estimation. Guarantees are distinguished into:

- **Real collateral**²: real estate, securities, pledges on movable assets, valued based on the expected market value at the time of realisation.
- **Personal guarantees**: sureties, letters of patronage, consortium counter-guarantees, which transfer recovery capacity to third parties.

²Collateral is an asset or set of assets pledged by a borrower to a lender as security for a credit exposure. In the event of the borrower's default, the lender has the right to seize, liquidate, or otherwise realize the collateral in order to recover, in whole or in part, the outstanding obligation.

A key indicator is the **Loan-to-Value (LTV)**:

$$LTV = \frac{EAD}{V_{\text{collateral}}}$$

where $V_{\text{collateral}}$ is the current value of the collateral. High LTV values indicate a smaller protection margin for the bank.

The operational definition explicitly distinguishes the effects attributable to *collateral* (real guarantees) from those stemming from *personal guarantees*. On the former, measurement incorporates prudential elements (i.e. asset-class-specific *haircuts*) and LTV dynamics; on the latter, the probability and severity of effective enforcement of the personal guarantee. This decomposition allows the loss-mitigation chain to be documented transparently and reconciled with recovery processes.

Discounting of cash flows

In *workout* estimation approaches, based on historical data of actual recoveries, recovery cash flows CF_t are discounted to the default date t_{DoD} with a discount factor DF_t consistent with internal policies, so as to correctly represent economic value:

$$RR = \frac{\sum_{t \geq t_{\text{DoD}}} CF_t \cdot DF_t - \text{costs}}{EAD}, \quad \text{whence} \quad LGD = 1 - RR.$$

The presence of *open* positions requires forward-looking estimates of residual recoveries up to the *Maximum Recovery Period*, maintaining consistency with historical evidence for homogeneous segments.

Prudential elements and long-run consistency

In line with regulatory expectations, the LGD estimate incorporates:

- **LGD ELBE (*Expected Loss Best Estimate*)**: estimate of expected loss on exposures already classified as impaired (*defaulted*), based on realistic assumptions and without prudential margins, used for the valuation of recoveries and for consistency with IFRS 9 accounting metrics;
- **LGD *downturn***: conditioning on adverse phases of the economic cycle, where more prudent than the long-run average, to reflect stressed conditions in collateral liquidation markets and in recovery timelines;
- **NPL disposal effects**: explicit consideration of the economic impacts of any disposals on non-performing portfolios, to ensure consistency between realized loss and estimated LGD.

2.2.2 Methods

Estimation of *Loss Given Default* (LGD) can follow three complementary approaches: *workout*, *market* and *structural*. They differ by the nature of the data used, scope of application and how costs and procedural risks are incorporated.

Workout approach (discounted historical recovery flows)

The *workout* approach uses historical recovery flows, discounted and net of administrative and procedural costs, estimated on homogeneous sets (by tool type, collateral type/quality, geographical area, industry sector). Distinct treatment is provided for *closed*, *substantially closed* and *open* positions, with estimation of *Time To Workout* and a *Maximum Recovery Period* to project remaining recoveries for unresolved positions.

Market approach (distressed prices / implied spreads)

The *market* approach infers the recovery rate from *prices of distressed tools* - that is, as a discount on the sale price (i.e. sales of non-performing exposures) - or from *implied spreads* observed on comparable tools. Price information succinctly embeds market participants' expectations about future recoveries and associated timing and costs, allowing an *RR* measure consistent with current market conditions.

Structural approach (grids and haircuts)

The *structural* approach determines LGD using *grids by asset class* and *collateral-specific haircuts* (i.e. discounts applied to the value of collateral), integrating *cost adjustments* and *legal risks* (validity and enforceability of the guarantee, realisation timelines). Grids enable consistent application across large portfolios, ensuring proportionality and traceability of underlying assumptions.

2.2.3 Downturn and prudential estimation

The *downturn* estimation of *Loss Given Default* (LGD) aims to reflect unfavorable macroeconomic and market conditions, producing values more conservative than historical averages. The methodological setup requires: (i) identification of the adverse period; (ii) measurement of the loss uplift observed in the adverse period; (iii) prudent projection of such uplift onto current estimates, maintaining consistency with segmentations and LGD calculation rules.

Identification of the downturn period

The downturn horizon must be a temporal subset $\mathcal{T}_\downarrow \subset \mathcal{T}$ representing adverse conditions (i.e. decline in collateral prices, lengthening of realisation times, increase in recovery costs). The selection is motivated by historical evidence on variables directly linked to loss severity.

Reference measures: long-run and *downturn*

For each homogeneous segment g (by product/collateral/area), compute:

$$\overline{\text{LGD}}_g^{\text{LRA}} = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \text{LGD}_{g,t}, \quad \overline{\text{LGD}}_g^\downarrow = \frac{1}{|\mathcal{T}_\downarrow|} \sum_{t \in \mathcal{T}_\downarrow} \text{LGD}_{g,t},$$

where $\text{LGD}_{g,t}$ is computed from exposures with discounted recoveries and net costs according to the defined perimeter. $\overline{\text{LGD}}_g^{\text{LRA}}$ is the long-run average (cyclical anchor), whereas $\overline{\text{LGD}}_g^\downarrow$ is the average conditional on adverse phases. The *downturn* estimate is accompanied by:

- sensitivity analyses to choices of $\mathcal{T}_\downarrow$ and to stressed assumptions on the drivers;
- comparisons by time cohorts and by segments, with traceability of sources and transformations;
- consistency checks against historical levels (avoiding underestimation) and with the intended managerial/regulatory use.

The overall framework ensures that resulting LGDs are *more conservative* than historical averages under adverse conditions and, at the same time, consistent and reproducible across portfolios.

Downturn uplift: additive and multiplicative approaches

The prudential add-on is summarized by an *uplift* factor:

$$u_g^\downarrow = \frac{\overline{\text{LGD}}_g^\downarrow}{\overline{\text{LGD}}_g^{\text{LRA}}} \quad (\text{multiplicative}), \quad \text{or} \quad a_g^\downarrow = \overline{\text{LGD}}_g^\downarrow - \overline{\text{LGD}}_g^{\text{LRA}} \quad (\text{additive}).$$

Application to current estimates $\widehat{\text{LGD}}_i^{\text{base}}$ (for position $i \in g$) follows one of the two conservative rules:

$$\widehat{\text{LGD}}_i^\downarrow = \min\{1, u_g^\downarrow \cdot \widehat{\text{LGD}}_i^{\text{base}}\} \quad \text{or} \quad \widehat{\text{LGD}}_i^\downarrow = \min\{1, \widehat{\text{LGD}}_i^{\text{base}} + a_g^\downarrow\},$$

with truncations to $[0, 1]$ and preserving the risk ordering within the segment.

Stress of severity drivers

Alternatively, or in addition to the historical uplift, a *stress* is applied to the drivers that determine severity:

$$\text{RR}^\downarrow = \frac{\sum_{t \geq t_{\text{DoD}}} (CF_t^\downarrow \cdot DF_t^\downarrow) - \text{costs}^\downarrow}{\text{EAD}}, \quad \widehat{\text{LGD}}^\downarrow = 1 - \text{RR}^\downarrow,$$

where $CF_t^\downarrow$ reflects higher collateral *haircuts* and realisation delays, $DF_t^\downarrow$ incorporates discount rates/spreads under adverse conditions and $\text{costs}^\downarrow$ represents a procedural burden consistent with stress phases.

Prudential combination

To ensure consistency with historical experience and the prudential objective, the final segment-level estimate adopts a maximization rule among relevant anchors:

$$\widehat{\text{LGD}}_g^* = \max\{\overline{\text{LGD}}_g^{\text{LRA}}, \overline{\text{LGD}}_g^\downarrow, \overline{\text{LGD}}_g^{\text{base}}\},$$

extended at the single-position level as $\widehat{\text{LGD}}_i^* = \max\{\widehat{\text{LGD}}_i^{\text{base}}, \widehat{\text{LGD}}_i^\downarrow\}$, while preserving the monotonicity of risk classes within each segment.

2.2.4 Calibration

The calibration of *Loss Given Default* (LGD) consists of aligning the raw model output (baseline estimate by position/segment) to *shape* and *level* constraints consistent with empirical evidence and prudential targets, while preserving the risk ordering (*rank-ordering*). In this framework, one typically uses **monotone/polynomial functions** for shape adjustment and **level alignment** procedures with respect to long-run or stress targets (LRA/DT), with goodness-of-fit assessed via the *distance between realized and predicted curves* and by monitoring discrimination (i.e. gAUC).

Shape adjustment: monotone functions

Let $\widehat{\text{LGD}}_g^{\text{base}}$ be the initial estimate and s_g an ordering (score/class). The shape adjustment seeks a *nondecreasing monotone* function $\phi(\cdot)$ such that the calibrated surface $\widehat{\text{LGD}}_g^{\text{form}} = \phi(\widehat{\text{LGD}}_g^{\text{base}})$ preserves rank-ordering and reduces local discrepancies:

$$\phi \in \mathcal{M} = \{\phi : \phi'(x) \geq 0 \ x \in [0, 1]\}, \quad \widehat{\text{LGD}}_g^{\text{form}} = \min\{1, \max\{0, \phi(\widehat{\text{LGD}}_g^{\text{base}})\}\}.$$

Operationally, the adjustment is applied to *homogeneous subsets* and *risk classes*, with monotonicity constraints to ensure economic-credit consistency and temporal stability.

Shape adjustment: polynomial functions with monotonicity constraint

When adopting a polynomial parameterization, estimate

$$\phi(x; \boldsymbol{\theta}) = \theta_0 + \theta_1 x + \theta_2 x^2 + \dots + \theta_k x^k,$$

subject to constraints ensuring $\phi'(x) \geq 0$ over the operating range (i.e. pointwise linear constraints on a grid or penalties imposing nonnegative derivatives). The goal is a smooth curve that corrects local irregularities while preserving monotonicity across risks.

Level alignment: LRA/DT target (*mass balance*)

Given $g = 1, \dots, G$ homogeneous classes/segments with counts N_g and exposures EAD_g and fixing a group *level target* (i.e. $\overline{\text{LGD}}_g^{\text{LRA}}$ or $\overline{\text{LGD}}_g^{\text{DT}}$), apply an affine transformation $\psi(x) = \alpha_g + \beta_g x$ to the shape surface, determining (α_g, β_g) to satisfy the *mass balance* constraint:

$$\frac{\sum_{i \in g} \psi(\widehat{\text{LGD}}_i^{\text{form}}) \cdot EAD_i}{\sum_{i \in g} EAD_i} \approx \begin{cases} \overline{\text{LGD}}_g^{\text{LRA}} & (\text{long-run anchoring}), \\ \overline{\text{LGD}}_g^{\text{DT}} & (\text{downturn anchoring}). \end{cases}$$

Anchoring may be carried out by *risk classes* (micro level) or by *segments* (macro level), while keeping truncations in $[0, 1]$ and global monotonicity.

Assessment of the goodness-of-fit of the adjustment

The “quality” of the calibration is verified by comparing *realized curves* (observed LGD by class/segment) with *predicted curves* (post-calibration). Distance indices between profiles are adopted (i.e. L_1/L_2 norms on grid points or EAD-weighted mean differences) and residual *discrimination* (i.e. gAUC) and its cycle variations are monitored to ensure that the adjustment does not deteriorate separation power.

Summary procedure

The practice for *homogeneous risk sets and classes* follows the steps below:

- baseline estimation and ordering by risk class;
- **shape** adjustment with a *monotone* or *polynomial* function ϕ that preserves rank-ordering;
- **level** alignment via ψ against the LRA or DT target (constraint of *mass balance*);
- verification of the distance between *realized* and *predicted* and discrimination check (i.e. gAUC);
- application of truncations and consistency checks (stability by segment/time).

This chain ensures methodological consistency, transparency of the link between historical data and calibrated levels and robustness to possible cyclical fluctuations.

2.2.5 Performance

Performance metrics in credit risk pursue two complementary objectives: (i) to verify **discriminatory power**, i.e. the correct ordering of positions by risk severity or loss; (ii) to measure **accuracy** of estimates, understood as the consistency between predicted and realized values on a homogeneous basis over time and by segment.

Discriminatory power

Discrimination assesses how well the model assigns, on average, higher LGD values to exposures that actually experience more severe losses. A summary indicator is the *generalized AUC* ($gAUC$), built on the *concordance* rate of pairs of positions:

$$C = \Pr(\widehat{LGD}_i > \widehat{LGD}_j \mid LGD_i^{\text{obs}} > LGD_j^{\text{obs}}), \quad gAUC = 2C - 1,$$

where $\widehat{LGD}$ is the model prediction and LGD^{obs} is the realized severity (on an EAD basis) over the same evaluation horizon. Values close to 1 indicate a strong ability to rank positions correctly by severity, whereas values close to 0 indicate no discrimination. $gAUC$ is computed on homogeneous subsets (tool type, collateral, geography) and, where relevant, with weights proportional to exposure EAD to emphasize ordering over economically most significant amounts.

Accuracy: consistency of level and profile

Accuracy is assessed by comparing, for homogeneous sets, predicted and realized LGD. Define classes $b = 1, \dots, B$ (i.e. by quantiles of $\widehat{LGD}$ or by product/collateral combinations) and the aggregates:

$$\widehat{LGD}_b = \frac{\sum_{i \in b} \widehat{LGD}_i \text{EAD}_i}{\sum_{i \in b} \text{EAD}_i}, \quad LGD_b^{\text{obs}} = \frac{\sum_{i \in b} \text{Loss}_i^{\text{obs}}}{\sum_{i \in b} \text{EAD}_i},$$

where $\text{Loss}_i^{\text{obs}}$ is the actual loss (computed on discounted recoveries and net costs) for unit i . Profile-level deviation indicators are:

$$\text{WAE} = \sum_{b=1}^B w_b |\widehat{LGD}_b - LGD_b^{\text{obs}}|, \quad \text{WSE} = \sum_{b=1}^B w_b (\widehat{LGD}_b - LGD_b^{\text{obs}})^2,$$

with proportional weights

$$w_b = \frac{\sum_{i \in b} \text{EAD}_i}{\sum_{b \in B} \sum_{i \in b} \text{EAD}_i}.$$

Level consistency is checked, both by classes/segments and at portfolio (aggregate) level:

$$\frac{\sum_i \widehat{LGD}_i \text{EAD}_i}{\sum_i \text{EAD}_i} \approx \frac{\sum_i \text{Loss}_i^{\text{obs}}}{\sum_i \text{EAD}_i},$$

requiring that the EAD-weighted average of predictions match the average realized severity within the same horizon and scope.

Complementary diagnostics

For fine-grained control of quality and temporal consistency:

- **Predicted–realized curves by class:** plot of the pair $(\widehat{LGD}_b, LGD_b^{\text{obs}})$ across classes b , useful to highlight systematic over/underestimation in specific severity regions.
- **Temporal consistency:** periodic replication of the metrics on *out-of-time* cohorts and by segment, so as to distinguish occasional deviations from structural drifts.

Joint interpretation of the metrics

A high $gAUC$ highlights good *rank-ordering* ability for severity, whereas low values of WAE/WSE indicate correct calibration at both the single-profile and portfolio level. Interpretation must therefore be joint: $gAUC$ measures *discrimination*, while the weighted errors measure *accuracy*; both dimensions are necessary to validate the quality and consistency of estimates across the entire application scope.

2.3 Estimation of EAD (Exposure at Default) and CCF

2.3.1 Operational definitions

The *Exposure at Default* (EAD) is the amount to which the bank is exposed when the counterparty defaults. It is used to calculate expected loss (EL) and, in a regulatory context, enters the calculation of RWA.

General principle

For a position i observed at the reference time t_0 , the exposure at the time of default $t_{DoD} > t_0$ is defined as:

$$EAD_i(t_{DoD}) = Drawn_i(t_{DoD}) + Ancillary\ items_i(t_{DoD}),$$

where *ancillary items* are those elements which, by managerial and contractual practice, increase the exposure upon default (i.e. installments/fees accrued up to t_{DoD}); such components are treated in a way consistent with internal accounting rules and with the measurement horizon.

Fixed-utilization facilities

For *fixed-utilization* products (i.e. mortgages/amortizing loans), the exposure at default coincides with the residual *drawn* amount at the time of default, net of repayments made before t_{DoD} :

$$EAD(t_{DoD}) = Outstanding\ principal(t_{DoD}) + Ancillary\ items(t_{DoD}),$$

subject to temporal consistency (*time-stamping*) between state variables and the default event.

Variable-utilization lines and Credit Conversion Factor (CCF)

For products with a *drawdown option* (credit lines, revolving lines, advances), EAD must also consider the *undrawn portion* that can potentially be activated before default. Operationally, a *Credit Conversion Factor* (CCF) is applied to the undrawn share of the commitment at time t_0 :

$$EAD(t_{DoD}) \approx \underbrace{Drawn(t_0)}_{\text{drawn}} + \underbrace{CCF \cdot (Granted(t_0) - Drawn(t_0))}_{\text{converted undrawn}} + Ancillary\ items(t_{DoD}),$$

where the CCF prudently summarizes the *expected draw* of the line between t_0 and t_{DoD} . The factor is estimated on homogeneous bases (by product, obligor segment, channel), with consistency of observation and performance windows relative to the PD horizon.

Temporal consistency and scope

The operational definitions require:

- **Temporal alignment:** the quantities entering EAD (drawn, granted, ancillary items) refer to times prior to or equal to t_{DoD} and are determined according to documented *cut-off* rules consistent with t_0 and aligned with the assessment horizon.
- **Homogeneous segmentation:** CCF estimation is performed on homogeneous subsets by technical form/product, so as to correctly reflect pre-default utilization behaviors.
- **Conservative treatment:** in the presence of uncertainty about residual drawdown, CCF incorporates a prudential margin, preserving the monotonicity of risk across product classes.

2.3.2 Estimation methodologies

Methodologies for estimating exposure at default in variable-usage facilities aim to quantify the *pre-default draw* of the undrawn portion, translating it into a *Credit Conversion Factor* (CCF) consistent with the reference horizon. One can distinguish **cohort** approaches and **fixed-horizon** (in line with regulatory practice) or **variable-horizon** approaches, all explicitly aimed at capturing utilization behaviors preceding the default event.

Cohort approach

Homogeneous cohorts of lines (by product, obligor segment, risk class, channel) are constructed, indexed by the observation time t_0 . For each cohort, the *average draw* between t_0 and the default event is measured and related to the undrawn portion at origination:

$$\widehat{\text{CCF}}_C = \frac{\sum_{i \in C} (U_i(t_{\text{DoD}}) - U_i(t_0))^+}{\sum_{i \in C} (A_i(t_0) - U_i(t_0))^+}, \quad x^+ = \max\{x, 0\},$$

where $U_i(t)$ is the drawn amount of line i and $A_i(t)$ the commitment. The truncation $(\cdot)^+$ avoids negative contributions in the presence of repayments, preserving the prudential interpretation of the factor. Using cohorts ensures comparability over time and allows $\widehat{\text{CCF}}$ to be computed for subsets consistent with observed behaviors.

Fixed horizon

In the fixed-horizon design, a time interval H is defined (i.e. $H = 12$ months, consistent with the PD horizon) and the *calendar-time* draw is estimated on exposures that default within H . The fixed-horizon cohort estimator is

$$\widehat{\text{CCF}}_C^{(H)} = \frac{\sum_{i \in C} (U_i(t_0 + H) - U_i(t_0))^+}{\sum_{i \in C} (A_i(t_0) - U_i(t_0))^+},$$

with *cut-off* rules to ensure information availability at $t_0 + H$. The fixed horizon guarantees methodological alignment with one-year risk measures and avoids spurious dependence on heterogeneous time-to-default across exposures.

Variable horizon

In the variable-horizon design, drawdown is conditioned on the actual time to default $\tau_i = t_{\text{DoD},i} - t_0$. One then evaluates the *expected increase* in utilization at default, denoted $h(\tau)$:

$$h(\tau) = \mathbb{E}[(U(t_0 + \tau) - U(t_0))^+ \mid \tau], \quad \widehat{\text{CCF}}^{(\text{var})} = \frac{\int h(\tau) d\mu(\tau)}{\mathbb{E}[(A(t_0) - U(t_0))^+]},$$

where μ is the empirical distribution of τ in the cohort. In practice, one operates with *bins* of τ (i.e. 0–3, 3–6, 6–12 months), estimating h for each bin and aggregating using frequency weights. The variable horizon is sensitive to last-mile dynamics (drawdowns concentrated close to default) and enables granular insights for operational policies.

Conditioning on drawdown behavior

Regardless of the design, analysis is conducted in segments reflecting behavioral differences: level of *drawn/granted* at origination $\rho = U(t_0)/A(t_0)$, relationship age, channel/product, presence of early-warning, etc. An operational example provides CCF *by bins of ρ* :

$$\widehat{\text{CCF}}^{(H)}(\rho \in B_k) = \frac{\sum_{i: \rho_i \in B_k} (U_i(t_0 + H) - U_i(t_0))^+}{\sum_{i: \rho_i \in B_k} (A_i(t_0) - U_i(t_0))^+}, \quad k = 1, \dots, K,$$

so as to capture the different residual drawdown margin as a function of the initial degree of utilization.

Integration into EAD calculations

The estimated CCF feeds into the construction of line exposure at t_{DoD} as $\text{EAD} \approx U(t_0) + \widehat{\text{CCF}} [A(t_0) - U(t_0)]^+$, maintaining consistency between observation window, segmentation and valuation horizon. Cohort and fixed/variable-horizon methodologies are therefore complementary tools to prudently quantify historically observed *pre-default drawdown*, with traceability of assumptions and calculation steps.

2.3.3 Parametric/semiparametric models

The objective of modeling the *Credit Conversion Factor* (CCF) is to estimate, in a consistent and conservative manner at the counterparty level, the share of the undrawn line at time t_0 that will turn into exposure before or at the time of default. Let

$$Y \equiv \frac{(U(t_0 + H) - U(t_0))^+}{(A(t_0) - U(t_0))^+} \in [0, 1],$$

be the *fractional realisation* of drawdown over a horizon H (fixed or variable) and let $\mathbf{x}$ denote the vector of covariates (product/segment, initial utilization ratio, relationship age, early warning, etc.). Estimating the CCF therefore requires models that respect the natural constraint $Y \in [0, 1]$, handle *zero inflation* (absence of drawdown) and allow flexible yet stable forms.

Parametric setup (fractional GLMs)

A direct approach consists in specifying the conditional mean $m(\mathbf{x}) = \mathbb{E}[Y \mid \mathbf{x}]$ via a *Generalized Linear Model* with a link function mapping $[0, 1]$ to $\mathbb{R}$:

$$g(m(\mathbf{x})) = \beta_0 + \boldsymbol{\beta}^\top \mathbf{x}, \quad g(\cdot) \in \{\text{logit}, \text{cloglog}, \text{probit}\},$$

so that the prediction $\hat{m}(\mathbf{x}) = g^{-1}(\beta_0 + \boldsymbol{\beta}^\top \mathbf{x})$ always lies in $[0, 1]$. Two specifications are particularly useful:

- **Fractional logit:** $g(p) = \log(\frac{p}{1-p})$, suited to data with full support $[0, 1]$ and an interpretation in terms of drawdown *odds*;
- **Complementary log–log (cloglog):** $g(p) = \log(-\log(1-p))$, useful when the distribution of Y is asymmetric and concentrated near 0.

Estimation proceeds by maximum likelihood $\hat{\boldsymbol{\beta}} = \arg \max_{\boldsymbol{\beta}} \sum_i \log f(y_i; g^{-1}(\eta_i))$ with $\eta_i = \beta_0 + \boldsymbol{\beta}^\top \mathbf{x}_i$, adopting distributional families compatible with fractional data (i.e. quasi-binomial³). The natural constraint $[0, 1]$ is guaranteed by the choice of link.

³A *quasi-binomial* model is used when regression is applied to proportion-type variables (fractional data, i.e. default rates or observed LGD). It generalizes binomial regression: in the *pure binomial* case, a variable $Y \sim \text{Bin}(n, p)$ has variance

$$\text{Var}(Y) = np(1-p).$$

In the *quasi-binomial* model, the mean p is retained but a dispersion factor ϕ is introduced to match the empirical variance:

$$\text{Var}(Y) = \phi np(1-p).$$

This accounts for *overdispersion* (empirical variance exceeding the theoretical one), which is common in proportional data. Such models fall within the class of *Generalized Linear Models* (GLMs).

Two-stage models (hurdle/zero-inflated)

When many observations are zero (no drawdown), *activation probability* and *conditional intensity* are separated:

$$\Pr(Y > 0 \mid \mathbf{x}) = \sigma(\alpha_0 + \boldsymbol{\alpha}^\top \mathbf{z}), \quad \mathbb{E}[Y \mid Y > 0, \mathbf{x}] = g^{-1}(\beta_0 + \boldsymbol{\beta}^\top \mathbf{x}),$$

where σ is the logistic function, $\mathbf{z} \subseteq \mathbf{x}$ and g^{-1} as above. The final prediction is $\hat{m}(\mathbf{x}) = \widehat{\Pr}(Y > 0 \mid \mathbf{x}) \cdot \widehat{\mathbb{E}}[Y \mid Y > 0, \mathbf{x}]$. This decoupling improves stability and interpretability when *drawdown* is a rare event but, if present, can be substantial.

Beta regression - parametric continuous on (0,1)

For outcomes $Y \in (0, 1)$ (any zeros handled via ϵ -shift or a discrete component), assume $Y \sim \text{Beta}(\mu\phi, (1 - \mu)\phi)$ with $\mu = \mathbb{E}[Y \mid \mathbf{x}]$ and $\phi > 0$ the *precision*:

$$g(\mu) = \beta_0 + \boldsymbol{\beta}^\top \mathbf{x}, \quad \ell(\boldsymbol{\beta}, \phi) = \sum_i \log \text{Beta}(y_i; \mu_i\phi, (1 - \mu_i)\phi).$$

Beta regression offers flexibility in the conditional variance and good adherence when the distribution is continuous and asymmetric.

Semiparametric settings (GAM/monotone smoothing)

To capture nonlinear relationships while maintaining control over complexity, *Generalized Additive Models* are employed:

$$g(m(\mathbf{x})) = \beta_0 + \sum_j f_j(x_j),$$

where f_j are smooth functions (i.e. penalized splines). If the economics of the problem suggest monotone relationships (i.e. CCF increases as the residual utilization margin rises), impose $f'_j(\cdot) \geq 0$ or ≤ 0 through monotone bases/shape penalties. Alternatively, *isotonic regression* on an ordering variable (i.e. the ratio $\rho = U(t_0)/A(t_0)$) ensures a nondecreasing/decreasing mean curve without specifying a rigid functional form.

Practical aspects: constraints, robustness and prudence

Regardless of the modeling class, the following operational principles apply:

- **Support constraint:** estimates are mapped to the interval $[0, 1]$ (via link function or truncation) to ensure economic consistency;
- **Regularization:** penalties (ridge, lasso, or smoothing) are introduced to curb variance and improve *out-of-time* stability;
- **Consistent segmentation:** estimation is carried out on homogeneous subsets (by product, obligor, or channel), in line with observed utilization behaviors;
- **Prudential margin:** in the presence of residual uncertainty, conservative *uplifts* are applied to forecasts $\hat{m}(\mathbf{x})$ to reflect adverse scenarios;
- **Evaluation:** performance is measured with metrics specific to fractional variables (absolute and squared errors weighted by exposure, calibration by risk classes) and validated on *out-of-time* cohorts.

2.3.4 Segmentation by business line

The segmentation of the estimation of *Exposure at Default* (EAD) and the *Credit Conversion Factor* (CCF) is designed to ensure *behavioral homogeneity* within each analysis subset. In particular, the document provides for a partition by **lines of business**, **products** and **obligor types**, so as to reflect line usage patterns, origination policies and comparable operational profiles.

Formal setup

Let $\mathcal{U}$ be the set of variable-utilization lines considered. Segmentation defines a partition $\{\mathcal{S}_g\}_{g=1}^G$ such that:

$$\mathcal{U} = \bigcup_{g=1}^G \mathcal{S}_g, \quad \mathcal{S}_g \cap \mathcal{S}_{g'} = \emptyset \ (g \neq g'),$$

where each segment g is identified by triplets (business line, product, obligor type) with possible additional operational constraints (i.e. channel, area). *Behavioral homogeneity* implies that, within $\mathcal{S}_g$, the profiles of *pre-default drawdown*, *utilization/commitment* and *management policy* are comparable, thereby stabilizing the CCF estimate.

Construction and consistency criteria

For each segment $\mathcal{S}_g$ operational rules are adopted to ensure statistical quality and comparability:

- **Minimum granularity:** count $N_g \geq N_{\min}$ and mass $\sum_{i \in \mathcal{S}_g} \text{EAD}_i \geq E_{\min}$ to support stable estimates of CCF and EAD.
- **Consistency of use:** uniform definitions of *commitment*, *utilization* and *ancillary items* and shared *time-stamping* rules within the segment.
- **Temporal stability:** periodic verification of *homogeneity* by comparing distributions (score/usage drivers) across consistent time cohorts; in the presence of evidence of *drift*, the segmentation is reassessed.

Segment-wise estimation and shape constraints

Given a segment $\mathcal{S}_g$, the *CCF* is estimated on a historical basis consistent with horizon H :

$$\widehat{\text{CCF}}_g = \frac{\sum_{i \in \mathcal{S}_g} (U_i(t_0 + H) - U_i(t_0))^+}{\sum_{i \in \mathcal{S}_g} (A_i(t_0) - U_i(t_0))^+},$$

with possible articulation into operational classes within the segment that preserve the expected ordering (monotonicity with respect to the residual utilization margin). Line EAD is derived consistently with the operational definition:

$$\widehat{\text{EAD}}_i \approx U_i(t_0) + \widehat{\text{CCF}}_g (A_i(t_0) - U_i(t_0))^+ + \text{ancillary items}, \quad i \in \mathcal{S}_g.$$

Pooling and aggregation rules

If a segment does not meet quality requirements (small sample size, excessive volatility), *pooling* rules are applied with segments contiguous in terms of business line/product/obligor base, while prioritizing behavioral homogeneity. The aggregation is documented and tested to ensure that the *bias* introduced is lower than the reduction in variance achieved.

Segmentation governance

Segmentation is subject to *periodic review* to capture any portfolio developments (new products, channels, policy changes). Changes are accompanied by:

- analysis of *impacts* on CCF and EAD (before and after) by business line and by product;
- checks of *comparability* over time (tracking migrations between segments in the event of a redesign);
- updates to the methodological documentation and quality controls.

2.3.5 Downturn calibration

The *downturn* calibration of *Exposure at Default* (EAD) and *Credit Conversion Factor* (CCF) aims to render estimates conservative during stress periods, incorporating the adverse effects historically observed on line usage behaviors and on portfolio operating conditions. The methodological framework entails: (i) identifying a time window $\mathcal{T}_\downarrow$ representative of adverse conditions; (ii) measuring the increase in drawdown relative to the long run; (iii) applying conservative rules that preserve consistency, monotonicity and natural truncations.

Identification of the adverse period and comparison metrics

For each homogeneous segment g (business line/product/obligor type) define:

$$\overline{\text{CCF}}_g^{\text{base}} = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \text{CCF}_{g,t}, \quad \overline{\text{CCF}}_g^\downarrow = \frac{1}{|\mathcal{T}_\downarrow|} \sum_{t \in \mathcal{T}_\downarrow} \text{CCF}_{g,t},$$

where $\mathcal{T}$ is a long-run reference horizon and $\mathcal{T}_\downarrow \subset \mathcal{T}$ identifies stress phases. The same definitions apply to EAD.

Downturn uplift: additive and multiplicative specifications

The prudential add-on by segment can be summarized by:

$$u_g^\downarrow = \frac{\overline{\text{CCF}}_g^\downarrow}{\overline{\text{CCF}}_g^{\text{base}}} \quad (\text{multiplicative}), \quad a_g^\downarrow = \overline{\text{CCF}}_g^\downarrow - \overline{\text{CCF}}_g^{\text{base}} \quad (\text{additive}).$$

Application at the position level $i \in g$, consistent with horizon H , proceeds as:

$$\widehat{\text{CCF}}_i^\downarrow = \min\{1, u_g^\downarrow \cdot \widehat{\text{CCF}}_i^{\text{base}}\} \quad \text{or} \quad \widehat{\text{CCF}}_i^\downarrow = \min\{1, \widehat{\text{CCF}}_i^{\text{base}} + a_g^\downarrow\},$$

while preserving the support $[0, 1]$ and monotonicity with respect to the segment's risk ordering.

Stressing behavioral drivers and consistency with EAD

As an alternative (or in addition) to the historical uplift, one may directly stress utilization dynamics:

$$\widehat{\text{EAD}}_i^\downarrow \approx U_i(t_0) + \widehat{\text{CCF}}_i^\downarrow (A_i(t_0) - U_i(t_0))^+ + \text{ancillary items}^\downarrow,$$

where ancillary items[↓] prudently reflect any increases in accrued fees/interest consistent with stress phases; the use of $\widehat{\text{CCF}}_i^\downarrow$ captures more intense drawdowns and/or drawdowns occurring closer to default as observed in $\mathcal{T}_\downarrow$.

Consistency constraints and mass balance per segment

The *downturn* adjustments are checked against consistency constraints on the segments:

$$\frac{\sum_{i \in g} \widehat{\text{EAD}}_i^\downarrow}{\sum_{i \in g} A_i(t_0)} \geq \frac{\sum_{i \in g} \widehat{\text{EAD}}_i^{\text{base}}}{\sum_{i \in g} A_i(t_0)}, \quad \frac{\sum_{i \in g} \widehat{\text{CCF}}_i^\downarrow (A_i(t_0) - U_i(t_0))^+}{\sum_{i \in g} (A_i(t_0) - U_i(t_0))^+} \approx \overline{\text{CCF}}_g^\downarrow,$$

so as to ensure that the average level is consistent with the historical stress experience and that intra-segment ordering is preserved.

Monotonicity and truncations

The surface $\widehat{\text{CCF}}^\downarrow(\cdot)$ must remain monotone with respect to variables that order the *residual drawdown margin* (i.e. the ratio $\rho = U/A$ or operational risk classes), avoiding reversals that would make managerial interpretation inconsistent. All quantities are truncated to $[0, 1]$ and documented by *bin/class*.

Validation and monitoring

The *downturn* calibration is subject to:

- **sensitivity analyses** on $\mathcal{T}_\downarrow$ and on the choice between additive/multiplicative uplift;
- **out-of-time checks** on subsequent cohorts to attest robustness;
- **profile comparisons** (predicted–realized curves of EAD/CCF by class) to identify any systematic over- or underestimation in the highest-risk deciles.

These checks consolidate the prudential nature of the framework, ensuring that estimates are appropriately strengthened during stress periods and consistent with historical drawdown behaviors.

2.4 Integration of parameters into capital and use test

2.4.1 Formulas and economic meaning

In this section, we coherently organize the fundamental relationships that link the risk parameters (PD, LGD, EAD) to the measure of expected loss (EL) and, through the regulatory coefficient K , to the determination of risk-weighted assets (RWA).

2.4.2 Expected loss - EL

Pointwise definition

$$\text{EL}_i = \text{PD}_i \times \text{LGD}_i \times \text{EAD}_i$$

The relation expresses the *mean* of the economic loss over a given horizon: the probability of incurring the default event (PD), times the severity conditional on the event (LGD), times the exposure at default (EAD). The formula is at the level of the i -th position and extends to the portfolio by summation.

Portfolio aggregation

For a portfolio $\mathcal{P}$ of exposures indexed by i :

$$\text{EL}_{\mathcal{P}} = \sum_{i \in \mathcal{P}} \text{PD}_i \text{LGD}_i \text{EAD}_i.$$

Additivity reflects the linearity of expected loss; in practice, aggregations by *segment* with exposure weights are used, ensuring consistency of horizon and scope across the different parameters.

2.4.3 Capital requirement and RWA

RWA

To define Risk-Weighted Assets (RWA), the regulation links the capital requirement per unit of exposure, namely K , to the exposure through the factor 12.5 (the inverse of 8%):

$$\text{RWA} = 12.5 \times K \times \text{EAD}.$$

By construction, the corresponding *minimum capital* is

$$\text{Minimum capital} = 8\% \times \text{RWA} = 8\% \times (12.5 \times K \times \text{EAD}) = K \times \text{EAD}.$$

This relationship can be extended to the portfolio level as follows:

$$\text{RWA}_{\mathcal{P}} = 12.5 \sum_{i \in \mathcal{P}} K_i \text{EAD}_i, \quad \text{Minimum capital}_{\mathcal{P}} = \sum_{i \in \mathcal{P}} K_i \text{EAD}_i.$$

Meaning of K

K represents the IRB capital requirement per unit exposure at a one-year at the 99.9% confidence level, *net of* expected loss (EL), including a possible maturity adjustment:

$$K = \left[\text{LGD} \cdot \Phi\left(\frac{1}{\sqrt{1-R}} \Phi^{-1}(\text{PD}) + \sqrt{\frac{R}{1-R}} \Phi^{-1}(0.999)\right) - \text{PD} \cdot \text{LGD} \right] \cdot \text{MA} \times 1.06,$$

$$R(\text{PD}) = 0.12 \cdot S + 0.24 \cdot (1 - S), \quad S = \frac{1 - e^{-50 \text{PD}}}{1 - e^{-50}},$$

$$\text{MA} = \frac{1 + (M - 2.5) b(\text{PD})}{1 - 1.5 b(\text{PD})}, \quad b(\text{PD}) = (0.11852 - 0.05478 \ln(\text{PD}))^2.$$

where:

- Φ is the standard normal cumulative distribution function and Φ^{-1} its inverse;
- PD denotes the probability of default and LGD the loss given default (typically the *downturn* LGD);
- R is the asset correlation (a decreasing function of PD);
- M is the effective maturity in years (usually between 1 and 5);
- MA is the maturity adjustment factor.

K increases with higher PD, LGD and M (via MA) and decreases as the correlation R declines. The term $-\text{PD} \cdot \text{LGD}$ removes the EL (covered by provisions), so that K covers only the UL.

Note: For retail portfolios (residential mortgages and other retail segments), specific $R(\text{PD})$ functions are applied and the maturity adjustment is generally not used. For CRR corporate/SME exposures, $R(\text{PD})$ also includes a size adjustment. Therefore the form without the size term is valid only under a large-firm assumption (size adjustment = 0).

RWA density and managerial interpretation

The *RWA density* on a set $\mathcal{S}$ is

$$\delta_{\text{RWA}}(\mathcal{S}) = \frac{\text{RWA}(\mathcal{S})}{\sum_{i \in \mathcal{S}} \text{EAD}_i} = 12.5 \times \frac{\sum_{i \in \mathcal{S}} K_i \text{EAD}_i}{\sum_{i \in \mathcal{S}} \text{EAD}_i},$$

and measures, as a weighted average, the *regulatory intensity* of capital absorbed per unit of exposure. For a given perimeter, changes in δ_{RWA} reflect changes in risk parameters (and thus in K) or in the portfolio mix. Changes in δ_{RWA} reflect (i) the *mix* of exposures (risk classes, collateral, maturities), (ii) *rebalancing* of portfolio parameters (i.e. lowering PD through selection, lowering LGD via collateral), (iii) *management actions* (securitisations/sales) that modify the distribution of $K \cdot \text{EAD}$ over the total. The indicator is therefore a *steering* tool: it allows assessment of the expected effect of commercial decisions and risk transfer on capital leverage.

2.4.4 Consistency between quantities and use test

Metric consistency

The two calculation chains

$$\text{EL} \Rightarrow \text{mean loss}, \quad K \Rightarrow \text{RWA} \Rightarrow \text{minimum capital},$$

are conceptually distinct and complementary: the first governs the *mean* of expected losses, the second quantifies the *capital* for prudential coverage consistent with the risk profiles summarized by K . In all cases, consistency of time horizon and scope across PD, LGD, EAD and K is required. EL provides the level of expected average loss for economic and accounting assessments, whereas RWA and the derived minimum capital ($K \times \text{EAD}$) inform regulatory and allocation choices. Finally, the *density* of RWA enables homogeneous comparisons across segments/portfolios, within a framework of traceability and methodological consistency.

Summary

Quantity	Formula	Meaning
Expected loss	$\text{EL} = \text{PD} \cdot \text{LGD} \cdot \text{EAD}$	Mean losses over the reference horizon
RWA	$\text{RWA} = 12.5 \times K \times \text{EAD}$	Regulatory base for minimum capital
Minimum capital	$0.08 \times \text{RWA} = K \times \text{EAD}$	Regulatory capital absorbed

2.4.5 Managerial uses

The integration of risk parameters (PD, LGD, EAD) into economic and regulatory measures enables consistent and traceable managerial applications. In particular:

- *risk-adjusted pricing* incorporates both expected loss and the cost of capital into price determination, ensuring consistency between commercial terms and risk profile;
- *capital allocation* is based on the average and marginal contributions of individual portfolios or positions to the overall regulatory requirement;
- *management of NPLs* (non-performing loans) uses risk parameters to assess ex-ante the sustainability of exposures and, ex-post, the effectiveness of recovery flows;
- *portfolio effects* are reflected in the density of RWA, guiding choices on asset composition and diversification.

Risk-adjusted pricing

Given a credit relationship with exposure EAD and a consistent measurement horizon, the *price* (rate or fee) incorporates:

$$\underbrace{\text{EL}}_{\text{expected loss}} = \text{PD} \cdot \text{LGD} \cdot \text{EAD}, \quad \underbrace{\text{CC}}_{\text{cost of capital}} = \underbrace{K \times \text{EAD}}_{\text{minimum capital}} \times c_{\text{cap}},$$

where c_{cap} is the unit cost of capital and K is the regulatory coefficient per unit of exposure. With OI operating/funding costs and π the target margin, a concise structure is:

$$\text{Price} = \frac{\text{EL} + \text{CC} + \text{OI} + \pi}{\text{EAD}},$$

which makes explicit the split between the expected loss component and the remuneration of the absorbed capital. The joint use of PD, LGD, EAD and K enables pricing comparability across segments and products, with traceability of risk drivers.

Capital allocation

For a portfolio $\mathcal{P}$, minimum capital is $\sum_{i \in \mathcal{P}} K_i \text{EAD}_i$. Allocation to individual relationships or segments can follow:

$$\text{average capital}_i = K_i \text{EAD}_i, \quad \text{marginal capital}_i \approx \frac{\partial}{\partial \lambda} \sum_j K_j(\lambda) \text{EAD}_j(\lambda),$$

where λ represents a small change in exposure. The use of *marginal* keys preserves additivity and captures composition effects; alternatively, *cluster* allocation schemes aggregate to homogeneous sub-portfolios with subsequent disaggregation proportional to EAD or EL.

Management of NPLs

In the processes for managing non-performing loans, risk parameters support both a *conditional on default* view and an *ex-ante* view of economic value. In particular, the expected recoverable value conditional on default is:

$$\text{Expected recoverable value conditional on default} = \text{EAD} \times (1 - \text{LGD}),$$

whereas, from an ex-ante perspective, the expected economic value also incorporates the probability of default:

$$\text{Expected economic value ex-ante} = \text{EAD} \times (1 - \text{PD} \cdot \text{LGD}).$$

Consistency between ex-ante estimates (at the time of classification) and ex-post evidence (recovery cash flows) enables effectiveness assessments of workout strategies (internal workout vs. sale, restructuring, collateral enforcement) and the measurement of capital requirement impacts through the subsequent updating of PD, LGD and EAD.

Integrated interpretation and use test

The systematic use of parameters across the three areas - pricing, capital and workout of non-performing exposures - constitutes an effective *use test*: PD, LGD and EAD support *pricing*, *allocation* and *workout* choices, while δ_{RWA} enables portfolio monitoring consistent with the regulatory framework. In this way, *circularity* between risk measurement and decision-making processes is ensured, with traceability of the links between parameter changes and economic-capital impacts.

2.5 Fine-tuning and validation

2.5.1 Discriminatory power

The validation of discriminatory power verifies that the model correctly ranks exposures by risk, using a consistent set of indicators and graphical tools: *ROC*, *AUC*, *CAP*, *AR*, decile default-rate analysis, confusion matrix and *cut-off* optimization. These elements are used jointly, including comparisons by segment, in order to attest the quality of the *rank-ordering* and its consistency in application.

Decile rates and confusion matrix

Decile rates compare, for equiproportional ranking classes, the observed frequency of defaults, allowing verification of rate monotonicity by class and the absence of local inversions. The *confusion matrix*, built at the operational threshold, provides a precision-oriented view to support the choice of the *cut-off* in line with the application objectives (regulatory and managerial).

Rank-ordering analysis

Validation includes *rank-ordering* checks by segment, with the aim of verifying that the risk ordering is preserved across the relevant dimensions (portfolios, products, classes). This reading complements the summary indicators (ROC/AUC, CAP/AR) and makes it possible to isolate any local criticalities.

2.5.2 Calibration accuracy

Calibration accuracy assesses the statistical consistency between *observed defaults* and *expected defaults* from the PD model, through formal tests that compare observed and expected frequencies over a fixed horizon. The tools considered are **binomial tests** (at the portfolio level) and **chi-square tests** (on classes/groups), applied to validation samples and, where available, *out-of-time*.

General setup

Consider a set $\mathcal{S}$ of N exposures, with estimated probabilities $\{\widehat{\text{PD}}_i\}_{i=1}^N$ for a horizon H . We denote by

$$D = \sum_{i=1}^N \mathbf{1}\{Y_i = 1\}, \quad E = \sum_{i=1}^N \widehat{\text{PD}}_i,$$

the *observed defaults* and the *expected defaults*, respectively. A summary calibration measure is the *calibration ratio*

$$\text{CR} = \frac{D}{E},$$

which ideally is close to one in the absence of systematic over/underestimation.

Binomial test (portfolio level)

The null hypothesis states that, as an approximation / under (near) homogeneity of PDs, the number of defaults follows a binomial law with expected average probability:

$$H_0 : D \sim \text{Bin}(N, \bar{p}), \quad \bar{p} = \frac{1}{N} \sum_{i=1}^N \widehat{\text{PD}}_i = \frac{E}{N}.$$

One uses the exact binomial *p-value* or, for large N , the normal approximation:

$$Z = \frac{D - N\bar{p}}{\sqrt{N\bar{p}(1 - \bar{p})}}, \quad p\text{-value} = 2(1 - \Phi(|Z|)).$$

If the *p-value* is below a critical threshold, typically 0.01 or 0.05, H_0 is rejected, indicating *level miscalibration* (defaults on average over/underestimated). A confidence interval for $\bar{p}$ (i.e. Clopper–Pearson) enables a threshold check; consistently, an interval for CR can be reported via the *delta method*:

$$\widehat{\text{CR}} \pm z_{1-\alpha/2} \sqrt{\frac{\text{Var}(D)}{E^2}} \approx \widehat{\text{CR}} \pm z_{1-\alpha/2} \sqrt{\frac{N\bar{p}(1 - \bar{p})}{E^2}}.$$

Chi-squared test on groups (profile consistency)

To assess calibration *by risk profile*, $\mathcal{S}$ is stratified into G groups (i.e. score deciles) with sizes N_g , expected mean $\bar{p}_g = \frac{1}{N_g} \sum_{i \in g} \widehat{\text{PD}}_i$, observed defaults $O_g = \sum_{i \in g} \mathbb{1}\{Y_i = 1\}$ and expected defaults $E_g = N_g \bar{p}_g$. The χ^2 test (Hosmer–Lemeshow type) adopts the statistic, in approximate form:

$$\chi_{\text{HL}}^2 = \sum_{g=1}^G \frac{(O_g - E_g)^2}{N_g \bar{p}_g (1 - \bar{p}_g)} \sim \chi_{G-2}^2 \text{ (approximate),}$$

where the degrees of freedom and the quality of the approximation depend on how groups are constructed. A small *p-value* signals systematic inconsistencies *in some risk regions* (local over/underestimation), even in the presence of good average calibration.

Exposure-weighted versions

If the managerial objective emphasizes amounts, *exposure-weighted* versions are applied (EAD weighted). Defining weights w_i proportional to EAD_i and weighted means $\bar{p}^{(w)} = \frac{\sum w_i \widehat{\text{PD}}_i}{\sum w_i}$, exposure-weighted (approximate) calibration tests are constructed by replacing $N\bar{p}$ and $N_g \bar{p}_g$ with $\sum_i w_i \widehat{\text{PD}}_i$ and $\sum_{i \in g} w_i \widehat{\text{PD}}_i$ and variances with $\sum w_i^2 \widehat{\text{PD}}_i (1 - \widehat{\text{PD}}_i)$ (consistent with the calibration objective).

2.5.3 Stability and representativeness

The stability and representativeness of the estimates are safeguarded through a monitoring framework that combines indicators of *population shift* (PSI), evidence of distributional *drift* and macroeconomic *stress tests*, so as to ensure that the levels and profiles predicted by the model remain consistent over time with the application population.

Objective of the oversight

The objective is twofold: (i) to ensure the **stability** of the *rank-ordering* and of risk levels over the cycle; (ii) to verify the statistical **representativeness** of current data with respect to the development/validation scope, preventing degradation due to structural changes in the portfolio or the macro environment.

Population Stability Index and drift signals

The *Population Stability Index* (PSI) is computed on key quantities (i.e. score, PD, or operational drivers) to compare the current population with the reference sample. The PSI provides a summary measure of distributional deviation; significant changes trigger detailed, segment-level analyses to identify the components driving the change (obligor *mix*, product, area, channel).

Distribution drift

Drift is detected through periodic comparisons between predicted and realized profiles on homogeneous cohorts, with readings:

- **level:** differences between expected and observed average rates across consistent sets;
- **profile:** deviations by classes/deciles along the ranking, useful for locating any inversions or distortions of the risk curve;
- **composition:** changes in the *mix* (lines of business/products/types of obligors) that impact the scope of application.

The combination of PSI (global signal) and class-level analysis (local signal) makes it possible to distinguish transient fluctuations from structural drifts.

Macroeconomic stress tests

Macroeconomic *stress tests* assess the resilience of the estimates conditional on adverse scenarios consistent with the economic cycle. The framework provides:

- **Scenario selection:** identification of adverse macro/sectoral trajectories (i.e. contraction in activity, interest-rate stress, decline in realizable values) relevant to the portfolio;
- **Parameter projection:** application of the scenarios to the risk surfaces to obtain forecast profiles under stress, consistent with the measurement horizon;
- **Baseline comparison:** assessment of deviations (level and shape) relative to the baseline case and verification of the impact on summary measures (expected loss, capital absorbed).

The result informs the forward-looking assessment of stability: a surface that remains ordered and consistent under stress scenarios is considered more robust.

Monitoring protocol

Monitoring is periodic and structured on three levels:

- **Population:** PSI on driver variables and segments, with operational thresholds for triggering analyses;
- **Performance:** profile-stability checks (predicted vs. realized by segment) and level consistency on *out-of-time* cohorts;
- **Perspective:** macro and sectoral stress exercises with assessment of impacts on economic-regulatory metrics.

The outcomes feed into the model *governance*, supporting decisions on maintenance, re-calibration, or revision of the application scope.

2.5.4 Adjustments (MoC)

The *Margins of Conservatism* (MoC) are prudential adjustments applied to risk parameters and/or derived quantities in order to compensate for statistical uncertainties, informational limitations and potential model instabilities over time. They are set out in regulatory and *data governance* frameworks and apply in particular when external/pool data are used or when application conditions deviate from the development/validation scope.

Breakdown of the MoC (A, B, C)

The *Margin of Conservatism* (MoC) is an adjustment applied to risk parameters (PD, LGD, EAD, CCF) to address three areas of methodological risk, distinguished as **MoC A**, **MoC B** and **MoC C**. The adjusted parameter is defined as:

$$\theta^{\text{MoC}} = \theta^{\text{base}} \oplus \text{MoC}^A \oplus \text{MoC}^B \oplus \text{MoC}^C,$$

where

- θ^{MoC} : final value of the risk parameter after applying the *Margin of Conservatism*. It represents the prudential estimate used in validation, risk management and regulatory capital calculations.

- θ^{base} : initial estimate of the risk parameter, calculated on the model development data (PD, LGD, EAD, CCF). It is the “unconserved” starting point, i.e. the best statistical estimate.
- MoC^A : add-on component related to **data representativeness**. It covers risks stemming from external/consortial sources or from population–mix shifts relative to development. It is often measured through indices such as the *Population Stability Index* (PSI) or distributional–shift tests.
- MoC^B : component reflecting **estimation and specification uncertainty**. It accounts for sample variance, data quality/completeness and the robustness of model assumptions. It is computed using confidence intervals (i.e. $z_{1-\alpha} \cdot \widehat{\text{SE}}$), sensitivity analyses and tests on alternative models.
- MoC^C : component linked to **cyclical and adverse scenarios**. It ensures that parameters reflect the impact of unfavorable macroeconomic conditions (*downturn* scenarios) and any transmission lags to observables. It is calculated as the expected difference between the parameter estimated under adverse conditions and that in normal conditions.
- $\oplus$: operator combining the MoC components with the base estimate. It may be additive ($\theta^{\text{MoC}} = \theta^{\text{base}} + \sum_k \text{MoC}^k$) or multiplicative ($\theta^{\text{MoC}} = \theta^{\text{base}} \cdot \prod_k (1 + \rho^k)$),⁴ depending on the nature of the parameter and on regulatory constraints.

MoC A - Data representativeness It addresses risks due to external/consortial sources and mix changes. Deviations between populations are measured through stability indices (PSI, KL), hypothesis tests and reweighting. The margin is calculated as:

$$\text{MoC}^A = \gamma_1 \cdot \text{PSI}_S + \gamma_2 \cdot \sum_{j=1}^p \omega_j \text{PSI}_{X_j} + \gamma_3 \cdot |\hat{\theta}_{\text{ref}} - \hat{\theta}_{\text{rw}}|.$$

where

- $\gamma_1, \gamma_2, \gamma_3$: calibration coefficients (scale weights) governing the impact of the different instability indicators. They are set at validation so as to reflect the relative importance of each contribution.
- PSI_S : *Population Stability Index* computed on the distribution of the model *score*. It measures the change in the score distribution between the reference and current periods. High values indicate a loss of population representativeness.
- PSI_{X_j} : *Population Stability Index* for predictor X_j . It captures any distributional changes in key drivers relative to the development period. The analysis is conducted over p explanatory variables.
- ω_j : weight associated with driver X_j , reflecting the variable’s importance in the model construction (i.e. statistical importance, predictive contribution, sensitivity).
- $\hat{\theta}_{\text{ref}}$: estimate of the risk parameter θ (PD, LGD, EAD, or CCF) computed on the reference population (i.e. the development sample).
- $\hat{\theta}_{\text{rw}}$: estimate of parameter θ recomputed on the current population via *reweighting*, i.e. adjusting the observed distribution to align it with the reference profile.
- $|\hat{\theta}_{\text{ref}} - \hat{\theta}_{\text{rw}}|$: absolute difference between the reference estimate and the *reweighted* estimate. It measures the magnitude of the deviation in terms of the expected level of the parameter, due to changes in the population.

⁴ ρ^k : the relative uplift associated with component k of the Margin of Conservatism ($k \in \{A, B, C\}$). It is expressed as a percentage increment (in decimal form) over the base estimate θ^{base} .

MoC B - Estimation and specification uncertainty It compensates for sample variance, coverage limitations, data quality and model specification/selection risk. In formulas:

$$\text{MoC}^B = \sqrt{(\text{MoC}_{\text{stat}}^B)^2 + (\text{MoC}_{\text{data}}^B)^2 + (\text{MoC}_{\text{spec}}^B)^2}.$$

where

- $\text{MoC}_{\text{stat}}^B$: component of the Margin of Conservatism that covers statistical estimation uncertainty due to sample variability. It introduces a prudential margin reflecting sampling error associated with the risk-parameter estimate.
- $\text{MoC}_{\text{data}}^B$: component of the Margin of Conservatism associated with **data quality and coverage**. It introduces a prudential margin to compensate for incomplete information or distortions in the development/validation dataset.
- $\text{MoC}_{\text{spec}}^B$: component of the Margin of Conservatism that covers **model specification risk**. It accounts for the sensitivity of results to the choice of functional form, inclusion/exclusion of predictors, or different estimation strategies.

The **statistical component**:

$$\text{MoC}_{\text{stat}}^B = z_{1-\alpha} \cdot \widehat{\text{SE}}(\hat{\theta}),$$

where

- $z_{1-\alpha}$: quantile of the standard normal distribution (mean 0, variance 1) corresponding to the confidence level $1 - \alpha$. For example:

$$z_{0.95} \approx 1.645, \quad z_{0.975} \approx 1.96,$$

for 95% and 97.5% confidence levels.

- $\hat{\theta}$: point estimate of the risk parameter (PD, LGD, EAD, CCF) obtained on development or validation data.
- $\widehat{\text{SE}}(\hat{\theta})$: standard error of the estimate, i.e. the square root of the estimated variance of $\hat{\theta}$. It measures how much the point estimate may vary due to the finiteness of the sample.

$\text{MoC}_{\text{stat}}^B$ therefore provides a margin equal to the width of the **upper half-confidence band** at level $1 - \alpha$. The larger the standard error or the higher the required confidence level, the greater the adjustment.

The **data-quality component**:

$$\text{MoC}_{\text{data}}^B = \lambda_1 \cdot \text{MR} + \lambda_2 \cdot \text{DQI}^{-1}.$$

where

- λ_1, λ_2 : calibration coefficients defining the relative weight of the two dimensions considered (missingness and overall data quality). They are set during internal validation and may vary by portfolio or segment.
- MR (*Missing Rate*): rate of informational gaps. It indicates the share of observations for which required data are unavailable or null/incoherent. It can be computed as:

$$\text{MR} = \frac{\text{number of missing or invalid fields}}{\text{total number of expected fields}}.$$

- **DQI** (*Data Quality Index*): composite index of dataset quality, normalized in $(0, 1]$, where values close to 1 indicate high quality (complete, consistent and timely data), while low values indicate severe qualitative deterioration. Using the inverse (DQI^{-1}) ensures that, as data quality worsens, the conservatism margin increases proportionally.

The formula balances two distinct yet complementary aspects: **completeness** of information (MR) and its **overall quality** (DQI), assigning each a calibrated weight (λ_1, λ_2) .

The **model-specification component**:

$$\text{MoC}_{\text{spec}}^B = \kappa \cdot \text{MAD}(\hat{\theta}^{(1)}, \dots, \hat{\theta}^{(M)}),$$

where

- $\hat{\theta}^{(m)}$ with $m = 1, \dots, M$: estimates of the risk parameter (PD, LGD, etc.) obtained from M alternative, data-plausible model specifications. Each $\hat{\theta}^{(m)}$ corresponds to a different configuration (i.e. subsets of variables, functional transformations, different regression techniques).
- $\text{MAD}(\hat{\theta}^{(1)}, \dots, \hat{\theta}^{(M)})$: *Median Absolute Deviation*, a robust measure of dispersion of the alternative estimates. It is defined as:

$$\text{MAD}(a_1, \dots, a_M) = \text{median}(|a_m - \text{median}(a_1, \dots, a_M)|).$$

High MAD values indicate that estimates are highly sensitive to model specification, signaling greater misspecification risk.

- κ : calibration coefficient, typically chosen to make MAD comparable to a robust standard deviation. A typical value is $\kappa \approx 1.4826$, which makes MAD equivalent to the standard deviation under normality.

MoC C - Cyclicity and adverse scenarios Ensures consistency with *downturn* estimates for LGD, CCF and EAD, including the severity, duration and lags of macroeconomic shocks. As an illustrative example, one may define:

$$\text{MoC}_{\text{macro}}^C = s(D) \cdot \Delta^\downarrow \theta, \quad s(D) = \min\left(1, \frac{D}{\overline{D}_{\text{crisi}}}\right).$$

where

- $\text{MoC}_{\text{macro}}^C = s(D) \cdot \Delta^\downarrow \theta$: MoC component reflecting the expected impact of the adverse macro scenario, appropriately scaled by the duration of the shock.
- D : observed duration of the adverse scenario (number of consecutive periods of economic stress).
- $\overline{D}_{\text{crisi}}$: average or typical duration of past crises used as a benchmark.
- $s(D) = \min\left(1, \frac{D}{\overline{D}_{\text{crisi}}}\right)$: scaling function that modulates the intensity of the margin based on the actual duration of the crisis, i.e. calibrates the impact on the basis of crisis length.
 - If $D < \overline{D}_{\text{crisi}}$, the margin is proportionally reduced.
 - If $D \geq \overline{D}_{\text{crisi}}$, saturation is assumed ($s(D) = 1$).
- $\Delta^\downarrow \theta$: difference between the parameter estimated under the adverse scenario and under the baseline. It is the **risk differential** induced by the macroeconomic cycle and quantifies how much the parameter increases (or worsens) during crises, i.e. it measures the deterioration of parameter θ under stress.

with

$$\Delta^\downarrow\theta = \theta^\downarrow - \theta^{\text{base}} = \mathbb{E}[\theta \mid Z = Z^\downarrow] - \mathbb{E}[\theta \mid Z = Z^{\text{base}}],$$

where

- θ : risk parameter under estimation, which may be PD, LGD, EAD or CCF.
- $\mathbb{E}[\theta \mid Z = Z^\downarrow]$: expected value of the parameter conditional on an adverse (*downturn*) macroeconomic scenario, characterized by worsening indicators such as unemployment, GDP and interest rates. It represents the prudential estimate of the parameter under stress.
- $\mathbb{E}[\theta \mid Z = Z^{\text{base}}]$: expected value of the parameter under the *baseline* scenario, i.e. under normal or long-run average economic conditions.

One may impose the constraint $\text{MoC}^C \geq \theta^\downarrow - \theta^{\text{base}}$, to ensure that the margin is not lower than the observed difference between the downturn and baseline estimates, thereby introducing a **prudential floor**.

Placement within the calculation chain

The MoC can be applied:

- *to the parameter*: $\text{PD}^{\text{MoC}}, \text{LGD}^{\text{MoC}}, \text{EAD}^{\text{MoC}}, \text{CCF}^{\text{MoC}}$, preserving monotonicity, truncations and temporal consistency;
- *to the derived measure*: i.e. on $\text{EL} = \text{PD} \cdot \text{LGD} \cdot \text{EAD}$ or along the stressed projection of EAD/CCF, documenting the marginal impact of the MoC relative to the base and downturn.

Interaction with downturn and stability

In the presence of *downturn* estimates (LGD/EAD/CCF), MoCs are applied so as to:

- not reduce the prudence embedded in the *downturn* measures;
- reflect the *severity* and *duration* of adverse phases and the observed transmission *lags*;
- maintain consistency with the outcomes on stability/representativeness (PSI, drift, macro stress).

Governance and review

The management of the *Margin of Conservatism* (MoC) is governed according to *change management* principles, in order to ensure transparency, traceability and consistency over time. In particular, the process provides for:

- **Review triggers**: activation of the process in the presence of regulatory updates, deterioration of model performance, significant changes in portfolio mix or credit policies, reduction in data quality or coverage, as well as in the event of structural breaks (*regime shifts*).
- **Impact assessment**: quantitative and qualitative analysis of the potential effects on the risk parameter and performance metrics, comparing the base estimate with the conservative estimate.
- **Approval process**: involvement of model-governance functions (development, independent validation, risk management and the competent decision-making committees), with a formal resolution on the introduction or update of the MoC.

- **Documentation and traceability:** retention of supporting evidence (analyses, datasets, tests) and assurance of result reproducibility, in line with audit and supervisory requirements.

2.5.5 Backtesting and challenger models

The *backtesting* and the structured comparison with *challenger models* constitute a central safeguard for the robustness and *governance* of the modeling framework. The objective is to verify ex-post the consistency between forecasts and observed outcomes, measure performance stability on independent cohorts and maintain methodological alternatives assessed under equivalent conditions.

Principles

The verification design is based on three operational principles:

- **Sample separation:** development (*development*), validation (*validation*) and *out-of-time* (OOT), to avoid dependencies between samples and to attest model generalization;
- **Iso-conditions:** comparisons (historical and against *challengers*) are conducted under equal data perimeter, time windows and *time-stamping* rules;
- **Multi-level reading:** *level* checks (predicted vs observed means) and *profile* checks (by classes/deciles), with breakdown by portfolios and segments.

PD backtesting: level and profile

Given a set $\mathcal{S}$ with estimated probabilities $\widehat{\text{PD}}_i$ over horizon H , the *expected defaults* $E = \sum_{i \in \mathcal{S}} \widehat{\text{PD}}_i$ are compared with the *observed defaults* $D = \sum_{i \in \mathcal{S}} \mathbb{1}\{Y_i = 1\}$.

- **Level:** consistency $D \approx E$ (reading of the *calibration ratio* $CR = D/E$, portfolio-level tests);
- **Profile:** across classes $g = 1, \dots, G$ (i.e. deciles), comparison of $E_g = \sum_{i \in g} \widehat{\text{PD}}_i$ vs $O_g = \sum_{i \in g} \mathbb{1}\{Y_i = 1\}$, with checks on monotonicity of rates and local deviations.

The analysis is repeated on OOT cohorts to attest stability over time.

LGD backtesting: realized severity

For loss severity, $\widehat{\text{LGD}}_i$ is compared with $\text{LGD}_i^{\text{obs}}$ (on an EAD basis, with recoveries discounted net of costs) within consistent operational classes (product/collateral/area). Summary indicators:

$$\text{WAE} = \sum_b w_b |\widehat{\text{LGD}}_b - \text{LGD}_b^{\text{obs}}|, \quad \text{WSE} = \sum_b w_b (\widehat{\text{LGD}}_b - \text{LGD}_b^{\text{obs}})^2,$$

where b are the validation classes and w_b are weights consistent with the exposed amounts. The joint level/profile reading makes it possible to identify systematic over/underestimation in specific severity regions.

EAD/CCF backtesting: pre-default drawdown

For variable-utilization lines, the undrawn portion converted under the model is compared with the *drawdown* observed over the reference horizon:

$$\widehat{\text{CCF}}_b \text{ vs } \text{CCF}_b^{\text{obs}},$$

where b are homogeneous segments (business line/product/obligor type). Consistency is assessed both in level and by profile, also verifying compliance with the $[0, 1]$ support constraints.

Performance indicators to support backtesting

Calibration checks are complemented by performance measures consistent with intended use:

- **Discriminatory power** (PD/LGD): assessment of the model's ability to correctly rank exposures by risk. The analysis is structured as follows:
 - *Rank-ordering*: comparison of the distribution of observed defaults along score or rating classes, verifying that riskier exposures exhibit higher default rates or higher average losses.
 - *Summary indicators*: use of aggregate discrimination measures such as:
 - AUC (Area Under the Curve) and ROC curve, for PD;
 - gAUC (generalized AUC) for LGD, based on concordance between expected and realized losses;
 - AR (Accuracy Ratio) and the CAP curve as complementary cumulative-read tools.
 - *Reading by segments/grades*: separate analysis by portfolio subsegments (i.e. retail, SME, corporate) or by rating grades, to highlight any differences in discriminatory power.
- **Stability**: *in-time* vs *out-of-time* comparison, with oversight on profile shifts (drift).

Taken together, these elements allow one to distinguish between one-off deviations and structural issues.

Challenger models: role and protocol

Challengers are alternative specifications assessed in parallel to the model in production (*champion*), for the purposes of continuous improvement and methodological oversight.

- **Iso-conditions**: same data windows, same pre-processing rules and identical verification perimeter (dev/val/OOT);
- **Comparison criteria**: comparative reading of calibration (level/profile) and performance (discriminatory power/stability), with documentation of trade-offs (*trade-off*);
- **Decision**: replacing the *champion* requires demonstrated superiority *and* OOT robustness, as well as consistency with uses (regulatory/managerial) and with traceability requirements.

The process is embedded in model *governance*, with outcomes and rationales archived and periodically reviewed.

Chapter 3

Governance

The governance of credit risk models is based on an integrated framework of processes, controls and responsibilities, organized into end-to-end phases: *data governance, development, implementation, testing, validation* (methodological and implementation), *risk management, compliance and steering and control functions* (Audit and Board of Directors). The separation of duties and the independence of controls ensure traceability, reproducibility and alignment with regulatory and internal requirements.

3.1 Data governance

3.1.1 Purpose

Ensure that the data used for the development and use of models are *fit for purpose*. In line with RDARR requirements/expectations and BCBS 239 principles, data governance must ensure the following data properties throughout the entire lifecycle:

- **Completeness:** ex-ante controls and ex-post verification of the percentage of records populated (including mandatory attributes and coverage by legal entity/business unit).
- **Accuracy:** ex-ante controls and ex-post verification of the percentage of records free from systematic errors and reconciled to authoritative sources (with defined tolerance thresholds).
- **Consistency:** ex-ante controls and ex-post verification of the percentage of records coherently transported and transformed *within* the same data source (intra-source coherence) and *across* different data sources (inter-source reconciliation), using common identifiers and taxonomies.
- **Timeliness:** ex-ante controls and ex-post verification of the percentage of records delivered within policy SLAs and update cadences, covering both punctuality of delivery and stability of production windows.
- **Uniqueness:** ex-ante controls and ex-post verification of the percentage of unique records and the absence of unjustified duplicates after joins, filters, or transformations.
- **Validity:** ex-ante controls and ex-post verification of the percentage of records conforming to the data dictionary, code lists and classification rules (domain/format checks and referential integrity).
- **Availability/Accessibility:** ex-ante controls and ex-post verification of the percentage of datasets accessible to intended stakeholders through authorised channels, within expected availability windows and with appropriate entitlements.

- **Traceability:** ex-ante controls and ex-post verification of the percentage of records with complete lineage (source system, processing steps/transformations, timestamps, owners) enabling reproducibility and auditability.

3.1.2 Organizational roles

- **Data owner** (line of business): responsible for the data governance process, for data correctness and for its availability within the domain.
- **Data steward** (data function): responsible for designing quality rules, first-/second-level controls, the glossary and the *data lineage*.
- **Data office/IT:** responsible for platforms, security, versioning and the management of separate environments (dev/test/prod).

3.1.3 Technical measures and controls

- **Data inventory and glossary:** mandatory metadata (field, definition, unit, domain, responsibility, transformations).
- **Quality and coverage:** the quality of the data used in the models is measured by means of summary indicators that take into account the main control dimensions (completeness, accuracy, coherence/consistency, timeliness, uniqueness). In particular:
 - **Data Quality Index (DQI)**

$$\text{DQI} = \sum_{h=1}^H \pi_h q_h,$$

where:

- $q_h \in [0, 1]$: normalized quality score with respect to dimension h . The dimensions considered include: completeness, accuracy, coherence, consistency, timeliness and uniqueness;
- π_h : weight assigned to dimension h , with $\pi_h \geq 0$ and $\sum_h \pi_h = 1$. It represents the relative importance of the dimension in constructing the composite index.

Values of DQI close to 1 indicate high-quality data across all dimensions, whereas low values signal material issues.

- **Missing Rate (MR)**

$$\text{MR} = \frac{\text{\#missing fields}}{\text{\#expected fields}},$$

measures data *completeness*. A low MR (close to 0) indicates that the information base is almost fully populated; high values require investigation of the causes (process errors, data collection issues, system inconsistencies).

- **Coverage Rate (COV)**

$$\text{COV} = \frac{\text{\#records with populated key attribute}}{\text{\#total records}},$$

quantifies the dataset *coverage* with respect to attributes deemed essential (i.e. obligor code, contract identifier, main collateral). A high value of COV ensures that critical data are present for almost all observations, guaranteeing representativeness and model robustness.

- **Data lineage:** tracking of data from their origin, through source systems and processing layers, to the modeling datasets. Formally, letting $T_1, \dots, T_L$ be the transformations applied to data $\mathcal{D}^{(0)}$, data lineage ensures the correct identification of all intermediate transformations such that: $\mathcal{D}^{(L)} = T_L \circ \dots \circ T_1(\mathcal{D}^{(0)})$.
- **Audit trail:** recording of operations performed on data and artifacts, including the outcomes of applied controls and any corrections (*data fixes*).

3.2 Development (modeling function)

3.2.1 Purpose

The modeling function is responsible for designing and estimating risk models, selecting predictors, defining algorithms and overseeing calibration and monitoring. It produces the methodological and technical documentation needed to ensure transparency, reproducibility and regulatory consistency. The development process follows a structured framework, starting from a broad set of candidate variables and proceeding through successive phases of selection and refinement. Therefore the development function explicitly defines:

- **Purpose of the model** (application scope, forecasting time horizon, quantities to be estimated such as PD, LGD, CCF).
- **Methodological assumptions** (statistical and operational hypotheses, functional and shape constraints, stability conditions, information regime).
- **Data sources** used and their acquisition methods, including representativeness and quality criteria.

All these elements are recorded in methodological and technical manuals, so as to ensure full reproducibility of results and traceability of decisions.

3.2.2 Deliverable

The development phase produces a set of mandatory documentary and technical artifacts, including:

- **Formal specification:** description of the purpose and scope of application, accompanied by a data dictionary with the description of the variables and their transformations, with an explicit statement of the adopted inclusion and exclusion rules.
- **Methodological documentation:** exposition of the theoretical assumptions, the methodological choices, including statistical and economic motivations, the model specifications, the objective functions, the selection criteria, as well as the rules for handling the variables.
- **Estimation protocol:** definition of random seed, samplings, segmentations into temporal cohorts, versions of the software libraries and parameterizations adopted, so as to guarantee independent replicability.
- **Output specifications:** schema of the fields returned by the model, admissibility ranges, numerical and categorical formats, error codes and rejection logs.
- **Calibration report:** summary of the calibration results with details of the assumptions and constraints applied, in addition to the testing results, accompanied by summary indicators (i.e. AUC, AR, WAE, WSE) and analysis by portfolio segment.

3.2.3 Development workflow

Model construction typically follows a process articulated in several phases:

- **Long list:** construction of a broad and redundant set of candidate variables, derived from heterogeneous information sources (demographic, credit, behavioral, macroeconomic data). The objective is to ensure maximum coverage of the possible determinants of risk, without yet performing any relevance selection.
- **Preliminary analysis of outliers and missing data:** identification and treatment of anomalies in the data, through *data cleaning* and imputation techniques. Variables with excessive levels of missing rate (MR) or distributions dominated by unrepresentative outliers are excluded or transformed.
- **Single Factor Analysis (SFA):** assessment of the statistical significance of each predictor entered individually in univariate models (i.e. single-factor logistic regressions). This phase makes it possible to estimate the actual individual predictive power, isolating weak or inconsistent variables.
- **Medium list:** definition of a reduced subset of variables that have passed the preliminary and univariate checks and that meet criteria of informational quality, temporal stability and forward-looking availability. This set constitutes the working basis for multivariate analyses.
- **Multifactor Analysis (MFA):** joint analysis of the selected predictors, in order to check for unwanted correlations, multicollinearity effects and relevant interactions. In this phase, metrics such as the *Variance Inflation Factor* (VIF), partial correlation tests and stepwise/penalized selection procedures (LASSO, elastic net) are used.
- **Short list:** construction of a narrow set of predictors deemed informative, robust and stable, that satisfy requirements of significance, temporal stability and future availability. The short list reduces the risk of collinearity and overspecification.
- **Choice of model specification:** definition of the model's functional form (i.e. logistic regression, generalized linear models, decision trees, hybrid models), of the variables actually included and of the transformations to apply (i.e. *binning*, scaling, monotonic transformations).
- **Integration of modules:** if present, merging of sub-scores (by product, segment, or counterparty) into a single score, through aggregation techniques or hierarchical models.
- **Rating classes:** discretization of the score or of the estimated probability of default (PD) into homogeneous, ordered classes, according to criteria of monotonicity and risk differentiation.
- **Calibration:** alignment of the estimated PD to a “target” default rate, derived from historical averages, forward-looking horizons, or regulatory requirements. The calibration phase entails defining the expected levels of the parameters, respecting shape and monotonicity constraints, as well as reconciling point-in-time and through-the-cycle estimates with the reference operational surfaces.

3.2.4 First-level controls

During the development phase, internal checks on consistency and completeness are performed:

- **Consistency** between the model's purpose, the available data and the usage requirements.

- **Completeness of the documentation produced**, with verification that all expected deliverables are present and up to date.
- **Traceability of the data flow**, i.e. the ability to reconstruct ex post all transformations undergone by the data, from the origin through to the input of the final model.
- **Verification of parameter stability** across time cohorts, residual diagnostics and specific tests for the risk of *data leakage* (improper use of future variables or proxies not permitted).
- **Sensitivity analysis of the estimates** with respect to data perturbations, specification changes, or exclusion of key variables (robustness). Targeted *stress tests* on critical predictors are included (for example, collateral, income, sector concentration), to assess the model's stability in extreme scenarios or those poorly represented in historical data.

These controls represent the first line of defense against model risk, anticipating subsequent independent validation checks.

3.3 Implementation

3.3.1 Objectives

The implementation phase aims to translate the methodological specifications into executable, controlled and documented software components. It must ensure equivalence between the theoretical model (*specification*) and its realisation in code (*implementation*), while also guaranteeing reproducibility, traceability and operational robustness. Release to production further requires the creation of formal deliverables, necessary to ensure correct execution, traceability and the ability to restore in the event of anomalies.

3.3.2 Deliverables

The main deliverables include:

- **Application package**, containing executable code, configurations and libraries required to run the model in the production environment.
- **Technical integration specification**, which describes the interfaces, information flows, input/output variables and infrastructure requirements for integration into enterprise systems.
- **Testing outcomes**, derived from functional and integration tests, including documented evidence comparing the development and production engines, with measures of result equivalence within predefined tolerance thresholds.
- **Release plan**, that defines the methods, timelines and responsibilities for the go-live, ensuring coordination with IT and risk management functions.
- **Roll-back safeguards**, i.e. formalized procedures to restore a previous version of the model in case of malfunctions or inconsistencies detected after release.

These deliverables are accompanied by checks of result consistency across environments (development, validation, implementation), alignment of variables and applied transformations, parameter monitoring and the provision of execution logs and *audit trail*, in order to ensure transparency and verifiability.

3.3.3 Technical safeguards and controls

- **Environment separation.** It is necessary to clearly distinguish the different stages of the application lifecycle:
 - **DEV** - development: environment used by quantitative and IT teams for code writing and initial testing;
 - **TEST**: environment dedicated to functional and performance testing, with controlled datasets;
 - **UAT** - *User Acceptance Testing*: environment where end users simulate real use cases to verify functional correctness and adherence to requirements;
 - **PROD** - production: final environment, subject to maximum segregation, with strict security and traceability constraints.

Such separation reduces the risk of errors, ensures cross-checks and prevents interference between development and operational use.

- **Versioning.** Each component (code, data, models, configurations) must be subject to *version control*, with tracking systems. Each release is accompanied by *tags* and *release notes* that document the changes introduced, backward compatibility and any *bug fixes* and *enhancements*. This makes it possible to reconstruct the model's history at any time and to ensure the reproducibility of analyses.
- **Input checks.** Before executing the model in the test or production environment, the input data must be verified through:
 - **Schema check**: conformity to the expected structure and types (i.e. numeric variables, categorical variables, dates).
 - **Range check**: consistency of numerical values with the expected domains.
 - **Consistency check**: plausibility checks across variables (i.e. the contract end date cannot precede the start date).
 - **Hard checks**: non-negotiable conditions that block processing if not met.
 - **Soft checks**: flexible conditions that generate *warning* and are tolerated within a predefined τ threshold (maximum percentage of allowed errors).

3.3.4 Methodological–implementational equivalence

To ensure that the implemented model (f_{impl}) is compliant with the methodological specifications (f_{spec}), the maximum difference between the two versions, computed on a representative set of test cases $\mathcal{S}$, must remain within a tolerance threshold ε :

$$\max_{x \in \mathcal{S}} |f_{\text{spec}}(x) - f_{\text{impl}}(x)| \leq \varepsilon.$$

- $f_{\text{spec}}(x)$: value of the parameter estimated by the theoretical model, according to the methodological specifications.
- $f_{\text{impl}}(x)$: value returned by the implemented code for the same input x .
- $\mathcal{S}$: set representative of the reference population (i.e. observations sampled across different segments, covering typical and tail cases).
- ε : maximum admissible tolerance threshold, defined based on numerical precision, the nature of the parameter and regulatory requirements.

In practical terms, the condition ensures that the software implementation is a faithful translation of the mathematical specifications, excluding discrepancies due to programming errors, differences in numerical precision, or inconsistencies across environments.

3.4 Model and process testing

3.4.1 Objectives

Testing activities include separate checks on development/validation (*dev/val*) samples and on temporal *out-of-time* samples, using indicators of both accuracy (consistency between estimated and observed values) and discriminatory power (ability to correctly rank exposures by risk). Checks are also performed on parameter stability over time and on the representativeness of the reference sample. The aim is to demonstrate that the implemented solution consistently meets the functional requirements (correctness of calculations and consistency with the methodological specifications) and the non-functional requirements (performance, robustness, operational stability). Tests constitute a fundamental pillar of *quality assurance* and must produce documented, traceable and reproducible evidence.

3.4.2 Types

- **Unit and integration tests.** They verify the correctness of individual software components (functions, modules, classes) and their integration within the end-to-end processing flow. The reference metric is the test pass rate:

$$TC = \frac{\text{\#tests passed}}{\text{\#total tests}},$$

which should approach 1 (coverage close to 100%). Unit tests are run on elementary functions, while integration tests verify end-to-end flow consistency.

- **Golden set.** This consists of a reference dataset whose expected outputs have been previously computed and validated as a *baseline*. The new version of the model must reproduce those outputs within predefined tolerance margins. The accuracy indicator is the maximum error:

$$ERR_{\infty} = \max_i |\hat{y}_i^{\text{old}} - \hat{y}_i^{\text{new}}|,$$

where $\hat{y}_i^{\text{old}}$ is the previously validated output and $\hat{y}_i^{\text{new}}$ is that computed by the new implementation. If $ERR_{\infty} \leq \varepsilon$ (with ε a defined threshold), the implementation is deemed equivalent.

- **Performance and resilience.** These assess the system's ability to meet non-functional requirements. Key metrics include:
 - **Latency** (average response time per request).
 - **Throughput** (number of processes per unit of time).
 - **Retry** rate (number of repetitions after a failure) and handled errors.
 - **Mean time to recovery** (MTTR) in case of failure.

These tests simulate extreme load conditions, component failures, network interruptions, or data anomalies.

- **Controls on data in production.** These monitor the quality and stability of the data processed during current model use. Two key indicators are:

- **Rejection rate**, i.e. the percentage of observations rejected due to schema, range, or logical consistency violations.
- **Input drift**, measured through distributional stability indices, for example the Population Stability Index (PSI) for each predictor X_j , computed between the reference period and the current period.

Anomalous PSI values or systematic deviations trigger *early warning* and potential *remediation* actions.

3.4.3 Expected outputs

At the end of testing, execution reports must be produced with the test outcome (passed/failed), details of the quantitative measures used (TC, ERR_∞ , latency, throughput, PSI), evidence of reproducibility (code, data, configurations) and traceability of anomalies together with a correction plan.

3.5 Validation

3.5.1 Structure

Model validation is independent, i.e. carried out by units not involved in development. Validation verifies methodological consistency, statistical robustness and adherence to regulatory requirements. Validation activities are organized into two distinct and complementary components:

- *Methodological validation*, performed by a function independent of development, with the objective of ensuring the model's theoretical and statistical soundness.
- *Implementation validation*, aimed at ensuring that the translation of the model into code and the IT infrastructure comply with the specifications and are secure and reliable.

This distinction reflects regulatory requirements for segregation of duties, avoiding conflicts of interest and ensuring independence in assessments.

3.5.2 Methodological validation

Objectives

Methodological validation is tasked with verifying:

- the model's **theoretical soundness**, namely consistency with regulation and with the relevant economic–financial principles;
- **statistical correctness**, with particular attention to specification assumptions, parameter identifiability and overfitting risks;
- **fitness for purpose**, i.e. the model's ability to meet regulatory (i.e. estimation of PD, LGD and CCF) and managerial (use test) objectives;
- **limitations and assumptions**, making explicit the domains of validity and potential sources of model risk.

Deliverable

Independent validation must produce at least:

- an **independent replication** of the estimates, with dataset and code separate from those used in development;
- a **benchmarking** against alternative specifications (competing models, simpler baselines);
- *out-of-time* (OOT) analyses to **test stability** on temporal cohorts not used in development;
- **sensitivity analyses** and **stress tests** on parameters and key variables, to assess robustness and risks of instability.

The outcome of validation is formalized in a summary assessment:

$$\text{Assessment} \in \{\text{Approved, Approved with conditions, Not approved}\}.$$

In the case of conditional approval or non-approval, **recommendations** must be issued, accompanied by an *action plan* specifying the responsible *owner*, implementation deadlines and the verification criteria for closing the actions.

3.5.3 Implementation validation

Technical safeguards and controls

- **Model–code equivalence.** Validation must ensure that the implemented version (f_{impl}) faithfully reproduces the methodological specifications (f_{spec}). The formal condition is:

$$\max_{x \in \mathcal{S}} |f_{\text{spec}}(x) - f_{\text{impl}}(x)| \leq \varepsilon,$$

where $\mathcal{S}$ is a representative test set and ε is the admissible tolerance threshold (i.e. differences due to numerical precision).

- **IT controls.** In addition to equivalence, implementation validation verifies:
 - **security** (authentication, authorizations, protection of sensitive data);
 - **environment segregation** (dev/test/prod);
 - **logging** and processing traceability;
 - the presence of **rollback** and **backup** procedures;
 - activation of application monitoring systems and **alerting**.
- **Minimum metrics.** Implementation validation relies on standard quantitative measures:
 - **run–time error rate:** $\text{ER} = \frac{\# \text{critical errors}}{\# \text{total runs}}$;
 - compliance with **SLAs** (Service Level Agreements);
 - **test pass rate**, as defined in the testing phase: $\text{TC} = \frac{\# \text{tests passed}}{\# \text{total tests}}$;
 - **code defect index:**

$$\text{DD} = \frac{\# \text{critical defects}}{\text{KLOC}},$$

where KLOC denotes the number of thousands of lines of code. High values of DD indicate low software quality and the need for refactoring.

3.5.4 Outputs

Validation - both methodological and implementation-level - must produce a formal report containing the assessment, recommendations, corrective actions and responsibilities, archived in official repositories with versioning and traceability.

3.6 Continuous monitoring

3.6.1 Purpose

Continuous monitoring aims to periodically verify the performance and adequacy of the model over time, ensuring that it remains consistent with the observed data, operating conditions and the regulatory and managerial purposes for which it was developed. The monitoring activity focuses both on the **level** (consistency between expected and observed values at the aggregate level) and on the **profile** (model behavior by classes and portfolio segments), ensuring continuity of use and the reliability of estimates. When issues arise, monitoring triggers review or re-calibration processes.

3.6.2 Deliverable

Monitoring produces systematic documentary evidence, including:

- **Periodic performance and stability reports**, which show trends in key metrics (discrimination, calibration, accuracy) and highlight any deteriorations.
- **Map of intervention triggers**, that is, a formalized set of thresholds and conditions that, when exceeded, require corrective actions or escalation (i.e. high PSI, AUC degradation, mix changes).
- **Corrective action plan**, indicating owners, deadlines and closure criteria, aimed at bringing the model back within acceptable thresholds.

3.6.3 Technical safeguards and controls

Monitoring relies on a set of quantitative and procedural tools:

- **Checks on *out-of-time* samples** to test the robustness of the model in periods not used during development, reducing overfitting risk.
- **Stability and drift checks**, with measurements on inputs (i.e. Population Stability Index on predictors), on estimated scores/PD/LGD and on observed outcomes, to detect distributional drifts and structural changes.
- **Oversight of alert thresholds**, by defining traffic-light bands (green, yellow and red) that promptly flag operational issues.
- **Traceability of outcomes and follow-up**, with archiving of reports, numerical evidence and decisions taken, in corporate repositories equipped with versioning and *audit trail*.

3.7 Risk management

3.7.1 Purpose

The *risk management* function is responsible for defining the requirements for model use, ensuring their integration into decision-making processes, setting operational thresholds and risk limits,

as well as overseeing monitoring metrics and *early warning* mechanisms. In this way, models are ensured not to be purely technical tools but to have an effective and traceable role in the corporate management of credit risk.

3.7.2 Technical safeguards and controls

- **Functional requirements.** Risk management ensures that the models are consistent with the processes of credit origination, exposure monitoring, price setting (pricing), calculation of regulatory capital (Pillar I) and internal capital (ICAAP), as well as with the determination of provisions (under IFRS 9).
- **Usage metrics and limits.** Indicators of actual use and operating limits are defined, which trigger escalation mechanisms. The main metrics include:
 - **Key risk indicators** (KRI), i.e. synthetic risk indicators on a portfolio or segment basis.
 - **Operational thresholds** expressed as traffic-light bands (green = regular use, yellow = enhanced monitoring, red = limit breach and escalation triggered).
 - **Early warning** indicators, such as unexpected changes in observed default rates or increases in drift in the estimated parameters.
- **Model inventory and model risk.** All models must be registered in a centralized inventory, with classification according to criteria of criticality, frequency of use, economic impact and degree of complexity. For each model a synthetic model risk score R is assigned, calculated as a weighted combination of model risk factors F_m :

$$R = \sum_{m=1}^M w_m s(F_m), \quad \sum_{m=1}^M w_m = 1,$$

where:

- F_m represents the observed value of the model risk factor (i.e. technical complexity, regulatory exposure, data criticality, frequency of use, obsolescence);
- $s(F_m)$ is a scaling function that normalizes the factor to a standard range (i.e. $[0,1]$);
- w_m is the relative weight assigned to the factor, based on its importance.

High values of R indicate models with a high risk profile, subject to more frequent monitoring and reinforced validation plans.

3.8 Internal audit

3.8.1 Purpose

The *internal audit* function provides an independent level of assurance over the overall model governance framework. Its mandate is to verify, with a risk-based approach:

- The **adequacy and effectiveness** of the **processes** that govern the life cycle of models (from development to retirement).
- Compliance with **corporate policies** and **external regulations**.
- The **proper segregation of duties** between development, validation, risk management and control functions.
- The **consistency between the approved models and the overall risk profile** assumed by the company.

3.8.2 Technical safeguards and controls

The audit function operates through periodic review plans, with:

- **Documentary tests** and on-site inspections.
- **Controls on the actual implementation of the recommendations** issued by validation and compliance.
- **Issuance of independent reports**, addressed to the Board of Directors and internal control committees.

Each review concludes with the formalization of recommendations and a remediation plan, tracked through to full closure, so as to ensure accountability and traceability.

3.9 Role of the Board of Directors and the CRO

3.9.1 Purpose

The Board of Directors approves the *model risk management* framework, the policies for development, use and validation, the *risk appetite* for model risk and the major adoptions/retirements of models. The CRO (Chief Risk Officer) ensures operational implementation, oversight of resources and the effective execution of approved plans and controls.

3.9.2 Reporting and decisions

- **Periodic reporting**: status of the model portfolio, based on *KRI* (key risk indicators) and *KPI* (key performance indicators), validation and audit outcomes, non-conformities and *remediation*.
- **Resolutions**: approvals for production release, material updates, justified waivers within predetermined limits, remediation plans.

3.10 Segregation of duties and RACI

3.10.1 Principle

The cornerstone of model governance is the **segregation of duties**, whereby no corporate function may *self-validate* or *self-approve* the activities it performs. This rule prevents conflicts of interest, strengthens the independence of controls and ensures the *traceability of responsibilities*. In other words, the party that develops a model cannot also validate it or approve its adoption, nor can those who govern its managerial use be the final oversight body at the same time.

3.10.2 RACI Matrix

To assign responsibilities clearly and in a documented manner, model lifecycle activities are mapped to a responsibility matrix according to the *RACI* framework:

- **R – Responsible**: the function *operationally responsible* for carrying out the activity (i.e. model development or code implementation). It performs the work, but does not necessarily approve the results.
- **A – Accountable**: the party *ultimately answerable* for the activity and the final decision. Each activity must have a single *Accountable*, typically at a managerial or governance level.

- **C – Consulted:** individuals or functions who, by virtue of their expertise, are consulted during the activity. This is a *supporting* responsibility involving two-way interaction (i.e. risk management providing requirements to development).
- **I – Informed:** functions that must simply be informed of the outcome of an activity or decision, to ensure transparency and organizational alignment. Interaction is one-way (i.e. reporting to the Board of Directors).

3.10.3 Objective

Building a RACI matrix ensures that every activity in the model lifecycle has a clearly identified *owner*; avoids overlaps and duplication of roles; ensures accountability in the event of errors or issues; and upholds the regulatory principles of independence and segregation of duties.

Activity	Data Gov.	Dev.	Impl.	Valid.	Risk Mgmt	Audit/BoD
Requirements & data definition	R	C	I	C	A	I
Model development	C	R	C	I	C	I
Implementation	C	C	R	C	I	I
Testing (unit/integr.)	C	R	R	C	I	I
Methodological validation	I	I	I	R	C	I
Implementation validation	I	I	C	R	C	I
Release to production	I	C	R	C	C	A
Usage monitoring	C	C	C	C	R	I
Audit and assurance	I	I	I	C	C	R/A

3.11 Change Management

Change management ensures that every modification to the models is preliminarily classified, authorized, implemented and tracked so as to preserve methodological consistency, regulatory compliance and the reproducibility of results.

3.11.1 Classification of changes

Changes are distinguished as:

- **Material:** they have a significant impact on parameters, outcomes, or the model’s scope of use. They require *prior approval* by the competent authority (ECB for SI / NCA for SLI) according to a formalized process. The main types of material changes include:
 - **Changes to the methodological specification** (i.e. addition/removal of predictors, change of estimation algorithm).
 - **Changes in the reference population** or in the application scope.
 - **Process changes** in implementation or in calculation systems.
 - **Correction of significant anomalies** identified during monitoring or audits.
 - **Updates required** by new regulations or regulatory guidelines.
- **Non-material:** technical or marginal in nature, they do not substantially alter results or application scope; they are subject to internal recording and control, without external authorization.

3.11.2 Authorization process for material changes

For material changes, a documented *authorization process* is in place, with defined timelines, including preparation of the change dossier, internal verification and submission to the Authority. Authorization is a prerequisite for release to production. The formal approval flow is structured in sequential phases:

trigger → impact analysis → testing and validation → approval → release → post-implementation review

3.11.3 Documentation

Versioning. Each phase of the change process must generate versioned evidence stored in official corporate repositories, accessible only to authorized parties. **Versioning:**

- Associates each release with a unique identifier and effective date.
- Retains details of the *methodological and operational changes*.
- Maintains the *version history* and the corresponding technical documentation, supporting transparency and reproducibility.

All evidence must be kept in archives equipped with an *audit trail*, enabling traceability of who made the change, when and for what reasons. This control ensures transparency, accountability and reproducibility - essential elements for regulatory compliance.

Changes made. Documentary evidence must include:

- **Data and datasets used** in development and validation, with the indication of sources and applied transformations.
- **Source code** and configurations, managed through *version control* systems.
- **Notebooks** or scripts containing analyses, tests and replicated estimates.
- **Test outcomes** and independent **validations**.
- **Minutes** of approval issued by the competent committees.

3.11.4 Lifecycle metrics

To monitor the effectiveness of the change management process, quantitative indicators are used:

$$\text{TTR} = \text{release date} - \text{request date},$$

which measures the **time-to-release**, i.e. the average time required to turn a change request into an operational release. Excessive values may signal organizational inefficiencies.

$$\text{Leakage} = \frac{\# \text{post-prod defects}}{\# \text{total defects}},$$

which quantifies **defect leakage**, namely the share of errors detected only after the production release. A high value indicates weaknesses in testing or validation phases.

$$\text{SLA compliance} = \frac{\# \text{tickets closed within SLA}}{\# \text{total tickets}},$$

which measures compliance with the **Service Level Agreement (SLA)**, i.e. the percentage of requests handled within the agreed maximum times. Values close to 1 indicate a good level of service.

Chapter 4

Regulation

4.1 European regulatory framework

The regulatory framework governing the development and use of internal models for credit risk in the European Union is based on a multi-level system of legal sources, structured around the *Capital Requirements Regulation* (CRR) and the *Capital Requirements Directive* (CRD), implementing the Basel principles. These rules are embedded within the three-Pillar structure of the supervisory framework, with cross-cutting effects on capital requirements, internal processes, model governance, data quality and public disclosure.

4.1.1 Three-Pillar Architecture

- **Pillar I (minimum requirements)** defines the criteria for calculating capital requirements for the main regulatory risks: credit, market and operational. For credit risk, the CRR allows the use of internal models (IRB) to estimate fundamental parameters such as probability of default (PD), loss given default (LGD) and exposure at default (EAD). The overall capital requirement can be expressed as:

$$K = K_{\text{credito}} + K_{\text{mercato}} + K_{\text{operativo}},$$

where:

- K is the total minimum capital requirement under the First Pillar;
- K_{credito} represents the capital requirement for **credit risk**, i.e. the possibility that a counterparty fails to fulfil its contractual obligations, it means $0.08 \cdot (\text{RWA}_{\text{credit}})$;
- K_{mercato} is the capital requirement for **market risk**, i.e. the risk of losses arising from adverse movements in market factors (rates, FX, spreads, prices), it means $0.08 \cdot (\text{RWA}_{\text{market}})$;
- $K_{\text{operativo}}$ covers **operational risk**, i.e. the risk of losses resulting from inadequate or failed internal processes, human error, IT systems, or external events, it means $0.08 \cdot (\text{RWA}_{\text{operational}})$.

Each term is calculated on the basis of risk-weighted assets (RWA):

$$\text{RWA} = \sum_{i=1}^n \text{EAD}_i \cdot \text{RW}_i,$$

and may be determined using different approaches, depending on the authorization granted by the supervisory authority and the institution's complexity. For example:

- for credit risk, one may adopt the standardised approach (SA) or internal ratings-based approaches (IRB);
- for market risk, either standard approaches or internal models approaches (*internal models approach*, IMA) may be used;
- for operational risk, starting from Basel III finalisation, the *Standardised Measurement Approach* (SMA) is mandatory, replacing prior approaches.

The total amount of required capital, expressed as K , is compared with the bank's available capital (Common Equity Tier 1, Tier 1 and Total Capital) to verify capital adequacy. Under the First Pillar, this capital must be at least 8% of risk-weighted assets (*Risk-Weighted Assets*, RWA), namely:

$$\frac{\text{Own Funds}}{\text{RWA}} \geq 8\%.$$

- **Pillar II (prudential review)** extends the regulatory framework beyond the First Pillar's minimum requirements, introducing a holistic assessment of institutions' capital and managerial soundness. In particular, it provides for:
 - the *Supervisory Review and Evaluation Process* (SREP), through which the supervisory authority assesses capital adequacy, risk governance, internal controls and the sustainability of the business model;
 - the *Internal Capital Adequacy Assessment Process* (ICAAP), whereby the bank must self-assess the capital needed to cover all relevant risks, including those not fully captured under the First Pillar (i.e. reputational, strategic, concentration risk);
 - the *Internal Liquidity Adequacy Assessment Process* (ILAAP), analogous to ICAAP but focused on the adequacy of the liquidity position.

Based on the outcome of SREP, the authority may impose:

- *Pillar 2 Requirements* (P2R): binding additional capital requirements, added to the First Pillar minimum;
- *Pillar 2 Guidance* (P2G): non-binding but monitored recommendations, intended to ensure an additional prudential margin in stress scenarios or where weaknesses are identified in the bank's risk profile.

In addition, the regulation requires holding an overall capital buffer (**Combined Buffer Requirement**, CBR), the amount of which is the sum of several regulatory components (Capital Conservation Buffer, Countercyclical Buffer, buffers for globally or domestically systemically important institutions, Systemic Risk Buffer). Failure to meet the CBR does not immediately constitute a breach of minimum capital requirements, but triggers progressive restrictions on dividends, remuneration and capital distributions, under the **maximum distributable amount** (MDA) rules.

The Second Pillar therefore strengthens the link between risk profile, internal governance and capital adequacy, fostering forward-looking and integrated risk management, in line with the principle of proportionality and the institution's long-term sustainability.

- **Pillar III (disclosure)** requires harmonized and comparable public disclosures on risks assumed, capital quality and the use of internal models. The CRR governs the contents, frequencies and formats of disclosure through Part Eight and the implementing technical standards (ITS). EBA/RTS standards specify publication templates and key metrics.

4.1.2 Operational and governance implications

The regulatory framework translates into operational obligations for institutions authorized to use IRB models. In particular, institutions are required to have:

- a **model governance** structure based on the segregation of duties among development, validation, approval and ongoing monitoring;
- robust **data governance** controls, including glossaries, traceability (*lineage*) and ex-ante and ex-post quality control;
- formalized **performance monitoring** processes, using statistical metrics (i.e. Gini, AR, Brier Score, PSI) and alert thresholds;
- a structured **model change management** framework, distinguishing between material and non-material changes, with traceability of rationales, approvals and notifications to the supervisory authority;
- alignment between regulatory measurements and accounting logic, especially where IFRS 9 ECL models are used.

Failure to comply with qualitative requirements may result in revocation of IRB authorization, a request to recalculate using the standardised approach, or the imposition of capital add-ons via P2R.

4.2 The Basel framework for banking regulation

4.2.1 Origins and objectives

The *Basel Committee on Banking Supervision* (BCBS) was established in 1974 by the central banks of the G10, in response to the failure of the German bank Herstatt, which highlighted the need for international cooperation to safeguard the stability of the global banking system. The BCBS operates under the aegis of the *Bank for International Settlements* (BIS) and is headquartered in Basel (Switzerland). The fundamental objectives of the Basel framework are:

- to ensure the **soundness and stability of the international financial system**;
- to promote the **harmonization of banking regulation** among member countries;
- to strengthen **prudent risk management** by intermediaries;
- to mitigate systemic risks and prevent banking crises of global scope.

This framework has evolved through a series of successive regulatory agreements, known as Basel I, II, III and IV.

4.2.2 Evolution of agreements

Basel I (1988). The first international accord aimed at establishing minimum capital requirements for banks. It introduced a *minimum solvency ratio* equal to 8% of risk-weighted assets (*Risk-Weighted Assets*, RWA):

$$\text{Capital Ratio} = \frac{\text{Own Funds}}{\text{RWA}} \geq 8\%.$$

Only *credit risk* was considered, with a standardised approach for the risk weighting of exposures. Though simple, the framework provided a uniform basis for international regulation.

Basel II (2004). It introduced a framework structured around three Pillars:

- **Pillar I:** minimum capital requirements for credit, market and operational risk;
- **Pillar II:** internal prudential review (ICAAP) and supervisory review and evaluation (SREP);
- **Pillar III:** transparency requirements and market discipline (disclosure).

The Basel II Accord maintains the overall minimum capital requirement of 8% of risk-weighted assets (*Risk-Weighted Assets*, RWA), but significantly refines the capital structure and the method of calculating both the numerator and the denominator of the *capital ratio*. Regulatory capital is divided into three categories:

- **Tier 1 Capital:** the highest-quality capital, comprising *Common Equity Tier 1* (common equity and reserves) and, in part, subordinated hybrid tools (preferred shares). It is considered the primary loss absorber under stress conditions;
- **Tier 2 Capital:** tools with lower loss-absorbing capacity, such as subordinated liabilities, unrealized gains and revaluation reserves;
- **Tier 3 Capital** (optional): usable exclusively to cover market risk.

The structure of the capital ratio can be formalized as:

$$\text{Capital Ratio} = \frac{\text{Own Funds}}{\text{RWA}} \geq 8\%,$$

where Own Funds = Tier 1 + Tier 2(+Tier 3) represents the available capital.

Basel II also introduces the option for banks to adopt advanced internal approaches (*Internal Ratings-Based*, IRB) for estimating risk parameters (PD, LGD, EAD), allowing greater risk sensitivity in the determination of RWA.

Basel III (2010). It introduced in response to the 2007–2008 financial crisis, it significantly strengthens both the *quantity* and the *quality* of regulatory capital required of banks. While maintaining the overall minimum requirement equal to 8% of risk-weighted assets (*Risk-Weighted Assets*, RWA), Basel III redefines the composition of the numerator of the *capital ratio*, introducing stricter explicit requirements and additional capital buffers. Regulatory capital is structured as follows:

- **Common Equity Tier 1 (CET1):** the highest-quality capital, consisting primarily of common shares, retained earnings and other tools with full loss-absorbing capacity. Basel III raises the explicit minimum CET1 requirement to 4.5% of RWA;
- **Additional Tier 1 (AT1):** subordinated tools with loss-absorbing features (i.e. contingent convertible bonds), which together with CET1 constitute *Tier 1 Capital* with a minimum requirement of 6.0% of RWA;
- **Tier 2 Capital:** lower-quality tools (subordinated liabilities and other reserves), contributing to the achievement of the overall 8.0% requirement of RWA.

In addition to these requirements, mandatory buffers composed entirely of CET1 are added:

$$\text{CBR} = \text{CCB} + \text{CCyB} + \text{G-SII/O-SII} + \text{SyRB},$$

where

- **CCB** is the **Capital Conservation Buffer**, intended to absorb losses in stress periods and ensure business continuity, set at 2.5% of risk-weighted assets (RWA);
- **CCyB** is the **Countercyclical Buffer**, intended to curb systemic risks accumulated during periods of excessive credit growth, varying by jurisdiction between 0 and 2.5% depending on the phase of the economic cycle;
- **G-SII/O-SII** is the buffer for globally or domestically systemically important institutions;
- **SyRB** is the **Systemic Risk Buffer**, designed to cover non-cyclical macroprudential risks.

The general formula for the capital ratio under Basel III can therefore be expressed as:

$$\text{Capital Ratio} = \frac{\text{CET1} + \text{AT1} + \text{Tier 2}}{\text{RWA}} \geq 8\% \quad (\text{plus additional CET1 buffers}).$$

Basel III also introduces the **Leverage Ratio**, to limit excessive bank leverage irrespective of the risk profile of exposures. It measures the ratio between Tier 1 capital and total exposure, which includes on-balance-sheet assets, derivatives, repurchase agreements and off-balance-sheet items, without applying risk weights:

$$\text{Leverage Ratio} = \frac{\text{Tier 1 Capital}}{\text{Total Exposure}} \geq 3\%,$$

Liquidity requirements: LCR and NSFR. An important innovation of Basel III is the introduction of quantitative constraints on the management of bank liquidity, in order to prevent liquidity crises similar to those that emerged during the 2007–2008 financial crisis. In this perspective, two complementary prudential requirements have been defined:

- the *Liquidity Coverage Ratio* (LCR), aimed at ensuring that the bank holds a sufficient amount of high-quality liquid assets to meet expected net cash outflows over a short-term horizon (30 days);
- the *Net Stable Funding Ratio* (NSFR), intended to ensure that the bank's funding structure is stable in the medium to long term, adequately covering assets with durable liabilities.

Liquidity Coverage Ratio (LCR). The *Liquidity Coverage Ratio* measures the bank's ability to withstand an acute, short-term liquidity stress scenario. The formula is as follows:

$$\text{LCR} = \frac{\text{High Quality Liquid Assets (HQLA)}}{\text{Net Cash Outflows over 30 days}} \geq 100\%.$$

where:

- HQLA: high-quality liquid assets (i.e. highly liquid sovereign securities, central bank reserves);
- Net Cash Outflows: expected net cash outflows over a 30-day horizon under a stress scenario.

Net Stable Funding Ratio (NSFR). The *Net Stable Funding Ratio* is a structural indicator that promotes a more resilient funding profile over time. It is defined as follows:

$$\text{NSFR} = \frac{\text{Available Stable Funding (ASF)}}{\text{Required Stable Funding (RSF)}} \geq 100\%.$$

where:

- ASF: the amount of funding deemed stable (i.e. stable retail deposits, long-term debt, equity);
- RSF: the amount of funding required to cover assets and off-balance-sheet exposures, weighted by their liquidity and maturity.

The NSFR requires that the amount of stable funding available be at least equal to the stable funding needed by the bank's assets over a one-year horizon (only items with a one-year maturity are assessed), limiting reliance on short-term volatile funding.

Basel III finalisation (2017–2022). With the finalization of Basel III and its European transposition into CRR II and CRR III, significant limits on the use of internal models have been introduced. The main measures include:

- **Input floors:** minimum thresholds on internally estimated parameters (i.e. LGD, PD) to avoid risk underestimation and promote comparability across banks.
- **Output floor:** a constraint that limits the RWA reduction effect due to the use of IRB models. Formally:

$$RWA_{\text{IRB effective}} \geq \alpha \cdot RWA_{\text{standard}},$$

where $\alpha = 72.5\%$ is the output floor level set by CRR III.

- **Methodological restrictions:** limitations on A-IRB for specific exposure classes (notably to financial institutions and large corporates above defined thresholds), requiring F-IRB or standardized approaches where mandated by CRR III/BCBS.

The objective of Basel III finalisation is to increase comparability across banks, limit model variability and strengthen market confidence.

4.3 European banking supervision

4.3.1 The Single Supervisory Mechanism (SSM)

The **Single Supervisory Mechanism** (SSM) is one of the core components of the *European Banking Union* and has been operational since 2014. Established by Regulation (EU) No. 1024/2013, it assigns to the **European Central Bank** (ECB) direct responsibility for the prudential supervision of *significant* banks in the euro area Member States, as well as of other European Union Member States that voluntarily decide to participate through close cooperation.

The main objective of the SSM is to strengthen the institutional framework of banking supervision within the Union, ensuring high, uniform and effective standards. In particular, the Mechanism aims to:

- ensure the soundness and resilience of the European banking system, reducing the risk of systemic crises;
- promote consistency and harmonization of prudential supervision, overcoming national disparities;
- contribute to the integration of the financial market and to the maintenance of overall financial stability in the euro area.

From an operational standpoint, the SSM is organized according to a two-level structure:

- the **direct supervision** exercised by the ECB over *significant banks* (*significant institutions*), identified according to size criteria, cross-border relevance, the extent of risks and economic significance;
- the **indirect supervision**, delegated to the national competent authorities (NCAs), which operate under the coordination of the ECB with regard to *less significant banks* (*less significant institutions*).

4.3.2 European implementation: CRR and CRD

The transposition of the Basel framework in the European Union takes place through two complementary legal tools:

- **Regulation (EU) No 575/2013** (*Capital Requirements Regulation*, CRR), which is directly applicable and binding in all Member States and sets out:
 - the definitions of regulatory capital (Common Equity Tier 1, Additional Tier 1, Tier 2), distinguishing between tools with full loss-absorbing capacity and lower-quality subordinated tools;
 - the calculation of minimum own funds requirements (*Pillar 1*) for credit, market and operational risk;
 - the *leverage ratio* and the liquidity framework: the *Liquidity Coverage Ratio* (LCR) is defined by Commission Delegated Regulation (EU) 2015/61 supplementing the CRR; the *Net Stable Funding Ratio* (NSFR) became binding with the *CRR II* package (Reg. (EU) 2019/876).
- **Directive 2013/36/EU** (*Capital Requirements Directive*, CRD), transposed into national legal systems, which lays down:
 - requirements on internal governance and the responsibilities of the management body;
 - the ICAAP (*Internal Capital Adequacy Assessment Process*) and ILAAP (*Internal Liquidity Adequacy Assessment Process*);
 - the **capital buffers** (CCoB, CCyB, O-SII/G-SII, SyRB) and the *supervisory review* framework (*Pillar 2*); *Pillar 3* disclosure obligations are set primarily in the CRR (Part Eight).

The most recent development is the **CRR III / CRD VI** package (finalisation of Basel III, often referred to as “Basel IV”), which includes restrictions on the use of advanced *IRB* models for exposures to *institutions* and *large corporates* (moving to *F-IRB* or the standardised approach), greater granularity in exposure classification and the phased introduction of the *output floor* from 2025 to 2030, with full application in 2030:

$$RWA_{\text{internal model}} \geq 72.5\% \cdot RWA_{\text{standardised}}.$$

The implementation of these rules is overseen by the *European Banking Authority* (EBA) and the *European Central Bank* (ECB) within the *Single Supervisory Mechanism* (SSM), to ensure consistent and rigorous application of prudential standards.

The regulatory framework: CRR, CRD and versions

European banking supervision rests on a structured regulatory framework that has progressively transposed the developments of the Basel Accords. The **CRR** governs the technical and quantitative aspects (own funds, RWA, leverage, liquidity), while the **CRD** sets out governance, internal processes and supervision.

Evolution of the CRR regulatory framework.

- **CRR I (2013)**: transposition of Basel III standards:
 - **Introduction of a coherent set of prudential requirements**, reflecting the content of the Basel III Accord: the structure of regulatory capital was defined and split into three components: *Common Equity Tier 1* (CET1), representing the highest quality capital; *Additional Tier 1* (AT1), consisting of subordinated tools with loss-absorbing features; and *Tier 2* (T2), comprising other eligible subordinated liabilities.
 - **Minimum own funds requirements** to be applied to the three main regulatory risks: credit risk, market risk and operational risk, using standardised or, where authorised, internal approaches.
 - **Leverage ratio**, i.e. a financial leverage coefficient measuring the ratio between Tier 1 capital and the institution's total exposure, irrespective of risk, in order to curb excessive balance-sheet expansion.
 - **Liquidity Coverage Ratio** (LCR) is specified by Commission Delegated Regulation (EU) 2015/61 supplementing the CRR and aims to ensure that banks hold sufficient liquid assets to withstand a 30-day liquidity stress.
- **CRR II (2019)**: post-crisis strengthening, including:
 - **Implementation of TLAC** (*Total Loss-Absorbing Capacity*) for **G-SII** (*Global Systemically Important Institutions*) and harmonisation of eligibility criteria for bail-in-able tools. TLAC is a minimum quantitative requirement in own funds and bail-in-able tools required of global systemically important banks. Introduced by the Financial Stability Board (FSB), TLAC aims to ensure that these banks maintain sufficient capacity to absorb losses in the event of failure without recourse to public support, preserving the continuity of critical functions.
 - **Proportionality principle for smaller banks**: regulatory *simplification* mechanisms for non-complex institutions (size thresholds and less burdensome reporting requirements), while preserving essential prudential safeguards.
 - **Net Stable Funding Ratio** (NSFR) becomes binding with the *CRR II* package (Reg. (EU) 2019/876) and imposes a structural balance between stable funding sources and longer-term uses.
- **CRR III (2025)**: completion of the transposition of Basel III finalisation:
 - Introduction of the **output floor** with a phase-in starting in 2025 and full application in 2030, limiting the benefits from internal models:

$$RWA_{\text{internal models}} \geq 72.5\% \cdot RWA_{\text{standardised}}.$$

- **Restrictions on the use of advanced IRB models** (*A-IRB*) for exposures to *institutions* and *large corporates*; for these portfolios, *F-IRB* or the Standardised Approach apply.
- **Structural revision of the requirements for operational and credit risk**:
 - *Credit risk (IRB)* - introduction of **input floors** on internal parameters to reduce unwarranted variability in requirements: minimum regulatory levels applied to **PD**, **LGD** (distinguished by type of collateral, i.e. unsecured vs secured) and **EAD** via floors on *credit conversion factors* (CCF) for certain off-balance-sheet exposures. The floors differ by exposure class and collateral scheme, with the aim of limiting excessive effects of internal modelling for the same economic risk.

- *Operational risk (SMA)* - elimination of the *AMA (Advanced Measurement Approach)*, based on internal models authorised by the supervisor) and adoption of a single standardised approach, the **Standardised Measurement Approach (SMA)**. The minimum requirement is determined as:

$$\text{ORC} = \text{ILM} \times \text{BIC}.$$

- ▷ **Business Indicator (BI)**: an accounting aggregate built from profit-and-loss items that represent exposure to operational risk (interest, lease and dividend components; services components; financial components).
- ▷ **Business Indicator Component (BIC)**: a piecewise function of BI with regulatory coefficients; in compact form,

$$\text{BIC} = \sum_j a_j \cdot \Delta BI_j,$$

where a_j are the marginal coefficients for each BI bucket.

- ▷ **Internal Loss Multiplier (ILM)**: a factor which, in the Basel standard, calibrates the charge based on historical operational losses.

Evolution of the CRD regulatory framework. The evolution of the Capital Requirements Directive (CRD) represents the qualitative complement to the CRR, governing internal governance, self-assessment processes and organisational safeguards. The main milestones are as follows:

- **CRD IV (2013)**: transposition of Basel III standards on the qualitative and organisational side. In particular:
 - **Introduction/strengthening of ICAAP and ILAAP**: the *Internal Capital Adequacy Assessment Process* (ICAAP) and the *Internal Liquidity Adequacy Assessment Process* (ILAAP) require banks to assess, autonomously and on an ongoing basis, the adequacy of capital and liquidity relative to their risk profile, beyond *Pillar 1* minimums, within the *Supervisory Review (Pillar 2)*.
 - **Governance and internal control requirements**: definition of the responsibilities of the management body and strengthening of *risk management, compliance* and *internal audit* safeguards, in line with the SREP process.
 - **Remuneration policies**: principles of *sound compensation practices* to avoid excessive incentives to take risk, consistent with the recommendations of the Financial Stability Board (FSB).
- **CRD V (2019)**: strengthening of the macroprudential framework and the supervisory role. The main innovations include:
 - **Clarification of the second pillar (Pillar 2)**: a formal distinction between binding requirements (**P2R**) and supervisory expectations (**P2G**), introducing greater transparency and proportionality in their application;
 - **Enhanced powers of competent authorities** in setting macroprudential buffers, imposing additional requirements and adopting measures of *early intervention*;
 - **Extension of supervision to banking holding companies and financial conglomerates**, through the introduction of the *Intermediate EU Parent Undertaking (IPU)* requirement for third-country groups, ensuring consolidated supervision at the European level;

- **Closer coordination between supervision and resolution**, aligning the framework with the **BRRD II** and the **Total Loss Absorbing Capacity (TLAC)** regime.
- **CRD VI (national transposition and application from 2026 with transitional phases)**: a directive complementary to the **CRR III**, which transposes on the qualitative side the finalisation of Basel III and introduces the explicit integration of **ESG risks** into the prudential framework. The main objectives are:
 - **Greater harmonisation of supervision** across the EU, establishing common standards for ICAAP, ILAAP and SREP, reducing national discrepancies and strengthening the EBA's role in supervisory convergence;
 - **Integration of ESG risks** (*Environmental, Social and Governance*) into governance structures, internal capital assessment and supervisory dialogue, making them an integral part of prudential risk management;
 - **Strengthening of macroprudential measures**, expanding the powers of national authorities and the ECB in managing systemic risks, including those related to the climate transition;
 - **Governance and control of internal models**, introducing uniform criteria for model development, validation and approval, in line with the *EBA Guidelines on Model Risk Management (EBA/GL/2024/04)*;
 - **Prudential regime for third-country branches**, introducing requirements for authorisation, reporting and cooperation among competent authorities to ensure consistent supervision within the EU single market.

The transparency obligations under *Pillar 3* are primarily governed by the **CRR** (Part Eight), while the **CRD** establishes the framework for *Pillar 2* (governance, SREP, ICAAP and ILAAP).

4.3.3 EBA guidelines and ECB expectations

The *European Banking Authority* (EBA) issues **Guidelines** that specify harmonized standards at the European level, including:

- qualitative and quantitative requirements for the **development and calibration of IRB models**;
- criteria for **independent validation**, through backtesting, benchmarking and stress testing;
- **data governance** and traceability requirements;
- **model governance** structures, with functional segregation between development, validation and approval.

The **ECB expectations** are consistent with the EBA guidelines, but with greater operational detail and prudential emphasis and are implemented through on-site or off-site reviews.

4.3.4 Focus: TRIM and LOM

TRIM – Targeted Review of Internal Models. The *Targeted Review of Internal Models* (TRIM) project, conducted by the **European Central Bank** between 2017 and 2021, represented one of the broadest supervisory initiatives ever carried out within the framework of the Single Supervisory Mechanism (SSM). The strategic objective was to ensure that the internal models used by banks to calculate capital requirements were compliant with regulatory requirements and methodologically robust. In particular, TRIM pursued three main objectives:

- **Harmonization of the use of internal models:** the ECB verified that modelling choices were consistent across comparable institutions, in order to reduce unwarranted disparities in the application of IRB models.
- **Reduction of unwarranted variability in capital requirements:** through a process of *on-site inspection* and in-depth technical analysis, TRIM identified areas of weakness that led to overly optimistic estimates of internal parameters (*PD*, *LGD*, *EAD*), affecting *Risk-Weighted Assets (RWA)*.
- **Strengthening methodological quality and governance:** banks were required to align their models with rigorous criteria for validation, documentation and traceability, according to common supervisory standards.

Overall, TRIM had a structural impact on internal modelling, contributing to regulatory convergence and to the credibility of prudential requirements based on IRB approaches.

LOM – Letters on Models (TRIM follow-up). *LOM letters* are the ECB’s ongoing follow-up after TRIM, with a view to **continuous supervision** over the quality of internal models. Unlike TRIM, which was extraordinary in nature and had a limited duration, LOM is an integral part of the authorities’ annual supervisory cycle. The main dimensions overseen by LOM include:

- **Validation of the robustness and adequacy of models over time,** verifying that predictive performance remains consistent with statistical assumptions and observed risk profiles;
- **Verification of consistency between models, portfolios and actual data,** through backtesting checks, *data quality* assessments and analyses of the significance of changes;
- **Benchmarking and cross-bank comparisons,** through comparative analyses that identify outliers and anomalies, including through the *EBA benchmarking* and *Supervisory Data* platforms.

LOM is a fundamental safeguard for the *regulatory use* of models, as it allows the ECB to request changes, usage restrictions, or revocations whenever significant deficiencies emerge.

4.3.5 ICAAP and ILAAP

In the context of the **Second Pillar** (Pillar II) of the Basel regulatory framework, two internal self-assessment processes represent essential tools for integrated risk management and for prudential planning at the corporate level:

ICAAP – Internal Capital Adequacy Assessment Process. ICAAP is the internal process through which each institution assesses its *capital adequacy*, taking into account the full set of risks to which it is exposed, its business model, the strategic planning horizon and macroeconomic conditions. The ICAAP is based on two fundamental dimensions:

- **Regulatory perspective (normative perspective):** verifies that the available regulatory capital (Total Capital) is sufficient to cover Pillar I requirements and any add-ons arising from the SREP process (*Pillar 2 Requirements* and *Pillar 2 Guidance*);
- **Economic perspective:** assesses the coverage of internal capital with respect to the *economically material risk*, measured according to internal models developed by the institution.

ILAAP – Internal Liquidity Adequacy Assessment Process. Similarly to the ICAAP, the ILAAP assesses the *adequacy of the liquidity position* of the bank against its liquidity risk profile, under both normal and stress conditions. The process covers:

- the availability of high-quality liquid assets (*High Quality Liquid Assets*, HQLA);
- the robustness of projected cash flows;
- sustainable funding strategies;
- emergency plans for liquidity crises (*Contingency Funding Plans*).

Both processes are documented annually and submitted to the supervisory authorities, who review them as part of the **Supervisory Review and Evaluation Process** (SREP). The effectiveness of the ICAAP and the ILAAP directly affects the SREP assessments, influencing the setting of additional capital requirements (*P2R*) or qualitative guidance (*P2G*), as well as requests for improvements to models or processes.

4.4 SREP

The **Supervisory Review and Evaluation Process** (SREP) is the prudential review process through which the supervisory authorities assess intermediaries' *capital* and *organizational* adequacy, with reference to their ability to measure, oversee and sustain the risks undertaken on an ongoing basis. The SREP provides the framework for the intermediary's *overall assessment*, within which the supervisory authority verifies: (i) the adequacy of capital to cover material risks and (ii) the soundness of the organizational set-up and risk governance processes, assessed in light of risk exposure. The *prudential* nature of the process aims to ensure operational continuity and resilience through the cycle. Following the analysis, the supervisory authority may impose:

- **Additional capital requirements (P2R):** further levels of regulatory capital, *add-ons* over and above the minima, to take account of the intermediary's specific risk profile.
- **Qualitative measures:** requests to strengthen processes and controls (organization, methods, operating practices) to safeguard proper risk management.

The combination of *quantitative* (P2R) and *qualitative* measures makes it possible to steer both the level of capital and the quality of internal safeguards.

4.4.1 SREP–ICAAP integration

The linkage between SREP and ICAAP is achieved by *aligning* the regulatory metrics (minimum and add-on requirements) and the *management* ones (internal risk/capital measures) within a unified information cycle, which includes the projection under stress of risk profiles and evidence of capital *headroom* in the face of adverse conditions. This integration ensures that the results of the ICAAP are *usable* in the supervisory process and that the SREP determinations translate into effective guidance for risk management.

4.5 Regulatory reporting and disclosure (Pillar 3)

Within the European regulatory framework, the supervisory authorities require banks to submit harmonized periodic information flows, in accordance with the technical specifications defined by the **EBA** (European Banking Authority). These flows fall within the **supervisory reporting** system and constitute essential tools for prudential monitoring and systemic analysis. The **COREP** and **FINREP** packages define the reporting standards to the supervisory authority, while **Pillar 3** governs public *quantitative* and *qualitative* disclosure on risks, capital and methodologies, with the aim of *transparency* and *comparability*.

4.5.1 COREP – Common Reporting

The **COREP** collects prudential data, providing a comprehensive overview of banks' risk profiles and regulatory requirements. In particular, it includes:

- the breakdown of the calculation of **own funds** and its components (CET1, AT1, T2);
- the **risk exposures** for credit, market and operational risk;
- the **minimum capital requirements** under the standardised or IRB approach;
- the **leverage ratio**, defined as:

$$\text{Leverage Ratio} = \frac{\text{Tier 1 Capital}}{\text{Total Exposure Measure}};$$

- the **liquidity indicators**, in particular the *Liquidity Coverage Ratio (LCR)* and the *Net Stable Funding Ratio (NSFR)*, required for monitoring resilience to short- and long-term liquidity stress.

4.5.2 FINREP – Financial Reporting

The **FINREP** standardizes accounting and financial disclosures, according to *IFRS*-based principles. It is mandatory for significant institutions subject to IFRS and includes:

- balance sheet and income statement, with granularity on assets, liabilities and margins;
- analysis of **non-performing exposures** (*Non-Performing Exposures*, NPE) and related provisions;
- details on **credit quality** and segmentation by regulatory portfolios;
- disclosure on **encumbered assets** and derivative liabilities.

The combination of COREP/FINREP enables cross-checking between regulatory measurements and accounting representation.

Technologies and usage

Both flows are processed in the **XBRL** (*eXtensible Business Reporting Language*) format and are transmitted to the competent authorities according to schedules governed by CRR and CRD. Supervisory authorities use these data to perform standardised **quantitative analyses** and cross-sectional checks (*peer review*), in addition to feeding European-level **benchmarking** and **stress test** exercises and detecting early signs of deterioration or systemic vulnerabilities.

4.5.3 Pillar 3: quantitative and qualitative disclosure

The **Third Pillar** disclosure requires the publication of *quantitative* elements (measures, quantities and indicators) and *qualitative* ones (methodologies, scopes and controls) relating to risks, capital and processes, with the aim of ensuring **transparency** and **comparability**. The disclosure makes explicit the link between the methodologies adopted and the results presented, reducing information asymmetries vis-à-vis authorities and the market.

4.5.4 Consistency and reconciliations

The information framework is structured to foster *consistency* between regulatory reporting flows (COREP, FINREP) and public Pillar 3 disclosure, promoting internal reconciliations between risk measurements and accounting figures and ensuring an aligned reading of the reported metrics. In this way, it supports effective *comparability* of information across institutions.

4.5.5 Organizational implications

The oversight of reporting and disclosure requires defined processes for roles, responsibilities and controls, with particular attention to data quality, traceability of transformations and timeliness of flows, in line with the established timelines. The combination of structured content, standardised formats and public disclosure enables effective governance of the regulatory scope.

4.6 Compliance and management of regulatory changes

Regulatory change management ensures the organization's timely and traceable alignment with external requirements, through a structured process that includes *monitoring of sources*, *gap analysis* and *planning and implementation of adjustments*. The process is documented and subject to defined accountabilities, thereby ensuring methodological consistency, transparency and reproducibility of outcomes.

4.6.1 Source monitoring

The framework provides a system of continuous **monitoring** of regulatory and supervisory sources, including the ECB, EBA and other relevant authorities. The monitoring scope covers new acts, updates, interpretative clarifications and supervisory expectations. The items identified are catalogued by *impact risk*, *urgency* and *scope of application*, establishing a structured register that enables their traceability throughout the entire management lifecycle.

4.6.2 Gap analysis

For each identified regulatory change, a **gap analysis** is performed as a systematic comparison between *current requirements* and *internal practices* across processes, methodologies, data and systems. The outcome identifies the *deviations* (gaps) and qualifies them by *materiality* and *criticality*, distinguishing between *process* (governance, controls), *methodological* (specifications/parameters) and *system* (IT implementations) adjustments. The gap analysis informs an *action plan* with responsibilities, timelines and acceptance criteria for the results.

4.6.3 Planning and implementation of the adjustments

Based on the identified gaps, an **adjustment plan** is defined and implemented that includes:

- *Processes*: updating organizational procedures, responsibilities and information flows;
- *Methodologies*: reviewing technical specifications and validation/monitoring safeguards to ensure alignment with requirements;
- *Systems*: adapting application engines and data quality/traceability controls, with *acceptance tests* and verification of consistency across environments.

Implementation is accompanied by updated **documentation** (methodological manuals, process diagrams, technical specifications) and by **outcome checks** that attest to full compliance and the operational effectiveness of the changes introduced.

4.6.4 Principles of governance, traceability and reporting

The entire cycle is governed by defined responsibilities (functions in charge of monitoring, analysis and implementation), **tracked** in change registers and **reported** through periodic reporting to the designated bodies, so as to ensure visibility over the state of compliance and the closure of gaps. Traceability covers the regulatory source, impact assessment, decisions taken, implementations carried out and verification of results.

Chapter 5

Accounting aspects (IFRS 9)

The accounting standard **IFRS 9** adopts a forward-looking measurement of *expected credit losses* (*Expected Credit Loss*, ECL), providing for estimates over a *12-month* or *lifetime* horizon depending on the deterioration status of the exposure. The logic is consistent with a forward-looking framework, in which the amount of the loss allowance reflects the expected risk over the relevant horizon and the tool's current credit quality.

5.1 Staging: classification into three stages

The *IFRS 9* framework provides for the classification of financial tools into **three stages**, determined by the degree of credit deterioration:

- **Stage 1:** credit risk has not increased significantly since initial recognition; the 12-month ECL applies.
- **Stage 2:** *significant increase* in credit risk since initial recognition; the *lifetime* ECL applies.
- **Stage 3:** *credit-impaired* (presence of objective evidence of loss); the *lifetime* ECL applies with treatments consistent with the impairment status.

The transition between stages occurs when conditions arise that indicate a worsening of the risk profile (*significant increase in credit risk*) or a *credit-impaired* status, resulting in a broadening of the loss estimation horizon.

5.2 ECL: definition and horizons

The **Expected Credit Loss** (ECL) is a probabilistic measure of credit risk, representing the expected value of losses on a credit exposure as a function of the probability of default and the expected loss in the event of default. In general terms, ECL is calculated as:

$$\text{ECL} = \sum_{t=1}^T \text{PD}_t \cdot \text{LGD}_t \cdot \text{EAD}_t \cdot D_t,$$

where:

- PD_t is the *probability of default* conditional on survival up to time t ;
- LGD_t is the expected *loss given default* at time t , i.e. the portion of exposure expected not to be recovered in the event of default;

- EAD_t is the expected *exposure at default* at time t , i.e. the amount of exposure expected at the time of default;
- $D_t = \frac{1}{(1+r)^t}$ is the discount factor, based on a discount rate r consistent with the asset's effective interest rate.

Depending on the **risk stage** assigned to the financial asset, under the three-stage model provided by *IFRS 9*, two main calculation horizons are distinguished:

- **12-month ECL** (*12-month ECL*): it is the *portion of lifetime ECL* that results from default events that are *possible within 12 months* after the reporting date (i.e., lifetime cash shortfalls generated by such events). It is calculated as:

$$ECL_{12-m} = PD_{12-m} \cdot LGD \cdot EAD \cdot D_1,$$

and applies to exposures classified in *Stage 1*, for which no significant increase in credit risk is observed since origination.

- **Lifetime ECL**: the expected loss over the entire remaining life of the financial tool, applying to tools classified in *Stage 2* (significant increase in credit risk) and *Stage 3* (credit-impaired or in default). Formally:

$$ECL_{\text{lifetime}} = \sum_{t=1}^T PD_t \cdot LGD_t \cdot EAD_t \cdot D_t,$$

where the time horizon $t = 1, \dots, T$ covers the tool's entire remaining contractual life.

For operational purposes, the ECL calculation requires the integration of:

- risk surfaces derived from the term structure of the probability of default (PD_t);
- forward-looking estimates of LGD and EAD, consistent with the forecast macroeconomic scenarios;
- the *staging* mechanisms, i.e. the transition between Stage 1, 2 and 3, governed by thresholds for the significance of credit risk deterioration;
- the discounting system consistent with the effective interest rate of the original contract.

5.3 Operational setup for ECL calculation

The construction of the *Expected Credit Loss* (ECL) requires a coherent and documented methodological framework that ensures alignment between risk assumptions and accounting requirements. In particular, it is necessary to preserve consistency between:

- **Estimation horizon**, which can be 12 months (*Stage 1*) or *lifetime* (*Stage 2 and 3*), depending on the deterioration in credit risk;
- **Information perimeter**, including internal historical data (observed default events, recovery rates) and, where relevant, *forward-looking* information consistent with the reference time horizon. Macroeconomic projections are incorporated through probability-weighted scenarios;
- **Segmentation and aggregation rules**, which ensure homogeneous grouping by risk profile, consistency in the modelling of parameters (PD, LGD, EAD) and transparency in the decomposition of loss components.

The entire framework is subject to reconciliation processes with credit risk metrics, particularly with the parameters used for regulatory purposes (i.e. IRB), in order to ensure consistency between accounting and prudential measurement. Furthermore, the operational setup must document in a traceable manner the information sources and methodological assumptions, ensuring **auditability** and **reproducibility** of accounting results.

5.4 Comparison between the IFRS 9 approach and the A-IRB model

The calculation of expected loss in the accounting framework (**IFRS 9**) and in the regulatory framework (**Advanced Internal Ratings-Based**, A-IRB) follows conceptually similar approaches, but is profoundly different in terms of objectives, horizons, parameters and purposes.

Aspect	IFRS 9	A-IRB (CRR)
Purpose	Accounting representation of expected loss in the financial statements	Prudential measurement of capital against credit risk
Horizon	Variable (12 months or <i>lifetime</i> , depending on the stage)	Fixed: 12 months, regardless of stage
Parameters	PD, LGD, EAD point-in-time , dependent on the economic cycle; incorporate <i>forward-looking</i>	PD, LGD, EAD through-the-cycle , estimated over long periods and more stable
Segmentation	Also on an accounting and managerial basis (i.e. business model)	Based on regulatory risk drivers
Scenarios	Multiple <i>macro</i> scenarios weighted by probability	TTC/LRA parameters with prudential components (i.e. downturn LGD), without mandatory probability-weighted forward-looking macro scenarios
Default	Entity-defined default notion, consistent with credit risk management practices and, in the EU context, generally aligned with the CRR/NDoD where relevant	Same NDoD definition, pursuant to Regulation (EU) no. 575/2013 (CRR)
Output	Accounting recognition of expected loss (<i>ECL</i>) in the financial statements	Capital requirement ($K_{\text{credito}} = \text{RWAs} \cdot 8\%$) for prudential coverage

In conclusion, while the IFRS 9 model aims at the *timely recognition* of expected losses, the A-IRB framework aims to ensure *continuous* and conservative prudential coverage of credit risk for solvency purposes. The structural differences require careful oversight of the reconciliation between the two domains, both at the validation stage and in reporting and disclosure.

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